g0-sense — Design Document

PCB-0015-A

hwde v1 pipeline

generated 2026-09-25 18:09:42

Contents

1 Board and Run Metadata **1**

2 Overview **1**

3 Requirements **2**

4 Architecture **4**

5 Schematic **26**

6 Layout **32**

7 Verification **32**

8 DFM and Fabrication **36**

9 Run Record (Appendix) **37**

10 Artifact Index **45**

1 Board and Run Metadata

Field	Value
board	g0-sense
part number	PCB-0015-A
workspace	`boards/PCB-0015-A_g0-sense`
phase	P10
gate status	all 5 recorded gates pass
state created	2026-08-27T06:35:38
state updated	2026-08-27T19:48:49

2 Overview

Brief: g0-sense (unattended container run, 2026-08-27)

A small 2-layer USB-C powered temperature/humidity sensor node - a product-scope board (no mode token: design normally), meant to be sent to JLCPCB as-is.

- Power: USB-C receptacle used for POWER ONLY (5 V VBUS; 5.1 k CC pull-downs so any USB-C source or a legacy A-to-C cable enables VBUS; D+/D- unconnected). Sensible input protection for a USB-powered device (VBUS TVS, resettable fuse or equivalent, reverse/inrush as the architect sees fit).
- 3.3 V rail from a fixed LDO, >= 300 mA, JLC Basic/Preferred part if one fits (AMS1117-3.3 class is fine), with proper input/output capacitors.

- MCU: STM32G030F6P6 (TSSOP-20) on the internal oscillator - no crystal. Decoupling per the datasheet, NRST with a push button, BOOT0 handled.
- Sensor: Sensirion SHT40 (or the SHT4x family member in JLC stock) on I2C with pull-ups; give it a thermal isolation slot/cutout or at least keep it away from the LDO and MCU heat.
- Interfaces: 4-pin SWD header (0.1 in), 4-pin UART header (0.1 in: GND, 3V3, TX, RX) for readout/logging, one Qwiic/STEMMA-QT connector (JST SH 1.0 mm 4-pin, GND/3V3/SDA/SCL) sharing the sensor I2C bus.
- One user LED (GPIO, series resistor) and one power LED.
- 2 layers, 1.6 mm FR-4, HASL, green; JLC economy PCBA for the SMD parts, prototype qty 5. Prefer JLC Basic parts; Extended parts only where the brief names the part (MCU, sensor, connectors).
- Size: whatever an honest layout wants, roughly 35 x 25 mm or smaller; four M2 mounting holes if they fit without hurting the layout.
- No battery, no mains, no USB data, no RF.

Fabrication spec snapshot

Key	Value
layers	2
width_mm	35.79
height_mm	28.34
qty	5
surface_finish	HASL
solder_mask_color	green
assembly	True

Stackup

Chosen stackup: `JLC2313_1.6` (reference/stackups.yaml, `available: true`)

3 Requirements

requirements - g0-sense

P0 requirements distilled from `brief/brief.md` (2026-08-27, unattended container run).

1. Function

Mode declaration: the brief carries NO mode token, so mode = none - design normally. Scope = product-scope per the brief's own words ("a product-scope board ... meant to be sent to JLCPCB as-is"): protection, filtering, connectors, thermal and enclosure-fit are all in scope, and their absence is a reviewer finding. Binding: the stated ˜35 x 25 mm size is a SOFT preference, not a hard cap (see section 5). Stage under study: none.

The board is a small 2-layer USB-C powered temperature/humidity sensor node. A USB-C receptacle supplies 5 V (power only, no data); a fixed 3.3 V LDO powers an STM32G030F6P6 MCU running on its internal oscillator, which reads a Sensirion SHT4x sensor over I2C. The same I2C bus is exposed on a Qwiic/STEMMA-QT connector; readout/logging is over a 4-pin UART header; programming/debug is over a 4-pin SWD header. One user LED and one power LED. Prototype quantity 5, JLC economy PCBA, meant to be ordered as-is.

2. Interfaces

- USB-C receptacle, POWER ONLY: 5 V VBUS input; 5.1 k pull-downs on CC1 and CC2 (so any USB-C source or a legacy A-to-C cable enables VBUS); D+/D- unconnected; no USB data.
- SWD header: 4-pin, 0.1 in pitch. ASSUMED: pins GND, 3V3, SWDIO, SWCLK (exact order per architect; brief states only "4-pin SWD header (0.1 in)").
- UART header: 4-pin, 0.1 in pitch: GND, 3V3, TX, RX - for readout/logging. ASSUMED: TX/RX are named from the MCU's perspective (TX = MCU transmit).
- Qwiic/STEMMA-QT connector: JST SH 1.0 mm 4-pin, GND/3V3/SDA/SCL, sharing the sensor I2C bus. I2C pull-ups required on the board (values per architect).
- User LED: one, driven from a GPIO through a series resistor.
- Power LED: one (which rail it indicates - VBUS or 3V3 - is per architect; the brief does not say).
- NRST push button (momentary reset).
- BOOT0: handled (method - strap, option bits, or jumper - per architect; the brief requires only that it is handled).
- Four M2 mounting holes, CONDITIONAL: only if they fit without hurting the layout (see section 5).

3. Power

- Input: USB-C VBUS, 5 V nominal, device draws < 1 A total. No battery, no charging, no mains.
- VBUS input protection required (product scope, brief-stated): TVS, resettable fuse or equivalent; reverse/inrush handling as the architect sees fit - the brief delegates the exact topology.
- 3.3 V rail: fixed LDO, >= 300 mA. JLC Basic/Preferred part preferred; AMS1117-3.3 class acceptable. Proper input/output capacitors per the chosen part's datasheet.
- Rail budget (GUESSES - to be replaced by the architect's numbers):
  - STM32G030F6P6 on internal oscillator: ~5-15 mA (guess)
  - SHT4x: < 1 mA average, single-digit mA peaks during measurement (guess)
  - LEDs: ~2-5 mA total (guess)
  - Qwiic downstream devices: not budgeted by the brief - see open question 2 (recommended default: reserve 100 mA).
  - On-board total is tens of mA; the brief's >= 300 mA LDO floor leaves the headroom for Qwiic loads.

4. Environment

Not stated in the brief. ASSUMED: indoor, room-ambient use (roughly 0-40 C), bare board, no ingress or vibration requirements (see open question 1). The one stated thermal/environmental requirement is layout-level: the SHT4x gets a thermal isolation slot/cutout, or at minimum is kept away from LDO and MCU heat, so self-heating does not corrupt the measurement.

5. Size & mounting

- Outline: "whatever an honest layout wants, roughly 35 x 25 mm or smaller". This is a SOFT preference - no HARD cap. It must NOT bind at P5 board_init; the layout earns the final outline, and ˜35 x 25 mm is a target to aim for, not a box to squeeze into (see open question 5).
- Mounting: four M2 mounting holes, CONDITIONAL - "if they fit without hurting the layout". The brief itself allows dropping or reducing them in favor of layout honesty; record that decision if taken.
- Height: no limit stated.

6. Quantity & budget

- Quantity: 5 boards, prototype run, assembled.
- Unit cost: no target stated (see open question 3). Cost posture from the brief: JLC economy PCBA; prefer JLC Basic parts; Extended parts only where the brief names the part (MCU, sensor, connectors).

7. Assembly

- JLC economy PCBA for the SMD parts.
- ASSUMED: single-sided SMD assembly (economy PCBA norm, and this part count does not need the second side).
- Through-hole parts: economy PCBA assembles SMD only, so the two 0.1 in headers are ASSUMED to ship unpopulated for owner hand-soldering (see open question 4).
- ASSUMED: NRST push button and the USB-C / JST SH connectors are SMD, JLC-assemblable variants where stock allows, so they ride the PCBA run.
- Board: 2 layers, 1.6 mm FR-4, HASL, green (brief-stated).

8. Compliance/safety flags

Each flag checked against the brief; none applies:

- Mains voltage: NOT APPLICABLE - brief states "no mains"; input is USB 5 V.
- Battery: NOT APPLICABLE - brief states "no battery"; no charging circuitry anywhere in scope.
- Motors: NOT APPLICABLE - no motor or actuator in the brief's scope.
- >30 V: NOT APPLICABLE - the highest voltage on the board is USB VBUS at 5 V nominal.
- High current (>3 A): NOT APPLICABLE - the brief describes a USB 5 V, < 1 A device.
- RF transmit: NOT APPLICABLE - brief states "no RF"; no radio, and no USB data either ("D+/D- unconnected").

9. Open questions

Closed-form, each with a recommended default. Unattended run: the orchestrator answers these on the owner's behalf and records the answers.

1. Operating environment: is indoor, room-ambient use (about 0-40 C, bare board, no ingress/vibration requirements) correct? Recommended default: YES. 2. Qwiic 3.3 V budget: how much current may external Qwiic/STEMMA-QT devices draw from the 3.3 V rail? Recommended default: reserve 100 mA (fits under the >= 300 mA LDO floor together with the on-board load). 3. Unit cost: is there a hard per-board cost cap? Recommended default: NO hard cap - minimize via economy PCBA and Basic parts as the brief already directs. 4. Header population: should the two 0.1 in headers (SWD, UART) ship unpopulated for hand-soldering, given economy PCBA assembles SMD only? Recommended default: YES - footprints and holes present, headers not populated at assembly. 5. Size confirmation: is ~35 x 25 mm truly a soft preference (the layout may exceed it if honesty demands), not an enclosure-driven hard cap? Recommended default: SOFT preference, per the brief's own wording ("whatever an honest layout wants").

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	unattended default: environment = indoor bench/room ambient, 0-40 C operating, bare board, no ingress or vibration rating	Q1 of requirements section 9; brief names no environment. Most conservative common-practice choice for a USB-powered indoor sensor node; parts will still be picked at industrial -40..85 C grade where that is the stocked option, so the assumption cannot make the board wrong.
P0	unattended default: Qwiic/STEMMA-QT downstream budget = 100 mA reserved on the 3V3 rail	Q2; brief does not budget it. 100 mA covers typical Qwiic sensor/display loads and still fits inside the brief's >= 300 mA LDO floor with margin for the on-board tens of mA, so it sizes the LDO and its thermals honestly without inflating the part.
P0	unattended default: no hard per-board cost cap; minimize cost via JLC economy PCBA and Basic parts as the brief already directs	Q3; brief states no budget. Cost is already bounded by the brief's own Basic-parts-preferred rule; inventing a cap could force a worse engineering choice.
P0	unattended default: the two 0.1 in headers (SWD, UART) ship UNPOPULATED - plated through-holes only, hand-solderable; only SMD parts go to JLC economy PCBA	Q4; brief says 'JLC economy PCBA for the SMD parts'. JLC economy assembly is SMT-only, so THT headers are BOM 'do not populate / customer supplied'. Standard practice for dev/sensor boards and it keeps the assembly quote in economy tier.
P0	unattended default: ~35 x 25 mm is a SOFT preference, not an enclosure-driven hard cap; no HARD outline cap binds at P5	Q5; the brief says 'whatever an honest layout wants, roughly 35 x 25 mm or smaller' and names no enclosure. Confirms requirements.md section 5; placement must not optimize to fit an unearned number (the bb-buck failure mode).

Phase	What	Why
P1	unattended default: LDO = AMS1117-3.3 class in SOT-223 (JLC Basic, C6186); governing thermal design point is the brief's literal >=300 mA rated case (0.51 W), not the ~150 mA realistic load	research/ldo.md escalated the SOT-223-vs-SOT-23-5 tradeoff; research/power.md resolved it with numbers: AMS1117 Tj 71-86 C at 40 C ambient and <=115 C even in the 500 mA abuse case, while AP2112K-class SOT-23-5 (theta-JA 184 C/W) hits Tj 134 C at the rated case with no copper fix available. The Qwiic port is an unfenced 3V3 output, so the rated case is reachable in the field. Basic part, 1.1M stock, $0.20.
P1	unattended default: a sub-6 V TVS clamping voltage is NOT a requirement; SMAJ5.0A/SMF5.0A-class (standoff 5.0-5.5 V, Vc ~9.2 V at Ipp) is accepted	research/protection.md flagged that no passive TVS clamps under 6 V. Nothing on VBUS is 6 V-rated: the only VBUS consumers are the LDO (AMS1117 Vin abs max 15 V), the input caps (16 V+ parts) and the PTC (16 V). Everything at 3.3 V sits behind the LDO. A ~9.2 V clamp during a surge is therefore inside every VBUS part's rating - the original <6 V framing was an over-tight derived spec, not a real one.
P1	unattended default: NO series reverse-polarity element (no Schottky, no P-FET ideal diode) on VBUS	Two independent reasons, both recorded in research: (a) a compliant USB-C cable cannot present reverse polarity (protection.md), and (b) research/power.md computed the end-to-end dropout budget and found a series Schottky's 125-190 mV would consume the entire AMS1117 headroom at the 4.75 V source corner (+0.11 V worst-corner margin as designed). The unidirectional TVS already crowbars a reverse fault. Adding it would make the board worse, not safer.
P1	unattended default: USB-C receptacle = 16-pin power-only part whose THT shield/mounting legs are accepted; JLC assembles it in the Economic PCBA tier	research/connectors.md correctly flagged that every 16P part in this class has THT shield legs and asked whether economy PCBA can place them. Verified directly on the JLCPCB part page for C165948 (TYPE-C-31-M-12): Assembly Process = SMT, Mount Type = Surface Mount Right Angle, and the part is listed as supported by BOTH Economic and Standard assembly. So the legs are not an assembly blocker. Part is JLC Extended - permitted, the brief names the connector.
P1	unattended default: SHT4x on-chip heater (75 mA peak) IS budgeted in the rail tree at <=10% duty	research/power.md's call, accepted: including it costs nothing (peak +3V3 still 185 mA, 240 mA with 30% headroom, under the 300 mA LDO floor and at 39% of the 500 mA CC-blind USB entitlement), while omitting it would turn a documented firmware feature into a hardware ECO later.
P1	unattended default: NRST tactile button - prefer the JLC Basic 5.1x5.1 mm part over the brief's smaller 3x2.5/4.2x3.2 mm size class if the layout has room; size class is a preference, Basic status and stock depth are the tiebreak	research/connectors.md Q2: the only Basic button is 5.1x5.1 mm (C318884, 962k stock, $0.02); the closest-size alternative carries a documented lib_pull symbol gotcha in this repo's own LEARNINGS. The brief never fixed the button size, and the board size is soft, so paying ~2.9 mm of edge for a Basic, gotcha-free part is the conservative trade. Architect may overrule at P2 if placement suffers.

Phase	What	Why
P1	unattended default: LED colours - user LED green, power LED red; both 0603, series resistors sized from the measured Vf band per colour	research/connectors.md Q3 left the assignment open. Red is the conventional power indicator and is the JLC Basic option with the low Vf band (1.8-2.4 V), which gives the largest resistor and the lowest current draw for an always-on LED. Green for the user LED is the conventional 'activity' colour; it is Extended in the KT-0603 family, which is acceptable at one extra unique part, and its 2.6-3.1 V Vf is fine from 3.3 V at a few mA.
P2	research task block-mcu-1 closed over 1 unverified draft record(s): mcu-boot-strap-only-when-option-bytes-consult-pin	–accept-drafts: the slot ships without this knowledge
P2	unattended default: BOOT0 gets a populated 10k pull-down on PA14 (pin 19, shared with SWCLK) - the board does NOT rely on the unverified factory-option-byte claim	The research record mcu-boot-strap-only-when-option-bytes-consult-pin was REFUTED by a fresh second reader: its load-bearing claim (factory 0xDFFFE1AA / nBOOT_SEL=1 makes the BOOT0 pin ignored) rests solely on a community.st.com forum page, and RM0454/AN2606 could not be acquired - st.com times out from this container (verified twice by curl, once by an agent). Rather than design on unverified knowledge, the board is made correct under EITHER option-byte state: a 10k pull-down guarantees BOOT0=0 at reset so the part boots main flash even if nBOOT_SEL=0. Cost is one 0402 resistor; 330 uA load on SWCLK is negligible to any debugger's push-pull driver, and the datasheet-confirmed internal pull-down on PA14 in debug mode already pulls the same way. This closes the gap without the source.
P2	unattended default: I2C pull-ups are 1.5 kOhm (not the 2.2 kOhm the P1 fragment proposed)	The draft record sht4x-i2c-pullup-bus-cap was refuted for recommending 2.2k over its own declared 150-200 pF bus budget: UM10204 Eq 1 at tr=300 ns gives Rp(max) = 354/Cb[pF] kOhm, i.e. 1.77k at 200 pF, so 2.2k needs Cb <= 161 pF and misses the Fast-mode rise time on a cabled Qwiic bus. 1.5k holds the ceiling to 236 pF and clears both floors (967 Ohm from UM10204 Table 10 at 3.3 V/3 mA, 390 Ohm from the SHT4x datasheet). Value independently recomputed by a second reader. architecture/constraints.json and the P4 schematic must use 1.5k.

Phase	What	Why
P3	unattended default: AMS1117 SOT-223 tab is treated as VOUT (+3V3) - the thermal pour is a +3V3 pour, not an isolated island	The datasheet-extractor honestly refused to assert it: the 'TAB IS OUTPUT' label on page 1 sits against the TO-252 view, and no text restates it for SOT-223. Resolved as tab=VOUT because (a) the datasheet's own 15 C/W junction-to-TAB figure means the tab is the die-attach pad, which on this family's leadframe is the VOUT lead, (b) the KiCad standard symbol/footprint pair the librarian pulls ties tab to VOUT, so schematic and layout agree by construction, and (c) it is the safe direction either way: if the tab were floating, a +3V3 pour on it is still harmless, whereas assuming 'floating' and isolating the copper would throw away the thermal design the 0.51 W design point depends on. Flagged for the schematic reviewer to confirm against the pulled symbol.
P3	unattended default: the SHT4x 20 V/ms VDD slew limit needs NO extra bulk capacitance - carried item CLOSED, not escalated	Two independent second readers read the SHT4x datasheet Table 4 slew row and quoted its condition verbatim: 'Voltage changes on the supply between VDD,min and VDD,max. Faster slew rates may lead to a reset' (min cell empty). That bounds supply changes DURING operation, not the cold-start ramp from 0 V - and a reset during power-on is the intended behaviour anyway (the part has a POR). In operation the 22 uF tantalum on +3V3 holds any load-step slew orders of magnitude below 20 V/ms. Adding bulk to slow a cold start would also fight the USB-C <=10 uF attach limit upstream. No hardware change; note it for bring-up.
P4	unattended default: WAIVE schematic-review W1 (no ESD array on the J2 Qwiic SDA/SCL) - ship the board without a TVS/ESD array on the I2C bus	Product-scope connector checklist allows meter-waived. Waived on four grounds: (a) the Qwiic/STEMMA-QT ecosystem norm (SparkFun/Adafruit reference boards) is a bare 4-pin JST SH with no array, so a user-facing part behaves as the ecosystem's users expect; (b) STM32G030 PA11/PA12 and the SHT40 both carry internal ESD structures and are 2 kV HBM rated, which is the standard handling rating for an indoor bench device; (c) the recorded environment is indoor bench/room ambient with no ingress rating, not a field-exposed port; (d) adding a 4-channel array now means a new Extended MPN plus setup fee, a fresh library pull and footprint verify, and re-opening a green schematic at P4 - real regression risk against a warning-severity, field-reliability-only exposure that cannot prevent bring-up. Recorded in reports/erc-waivers.md and shown at H2.

Phase	What	Why
P4	unattended default: CLOSE the carried verify item 'SHT4x VDD slew <= 20 V/ms vs AMS1117 startup ramp' (schematic-review W2) as accepted risk - no hardware change	Three findings close it. (1) PROVENANCE IS ON FILE, contrary to the reviewer's note: the 20 V/ms figure is research/records/sht4x-vdd-slew-power-up.yaml, sourced to research/sources/HT_DS_Datasheet_SHT4x.pdf p9 Table 4 and read visually TWICE (researcher + fresh second reader, maturity verified). The reviewer grepped parts/C2909890.json (the extraction JSON) and the LCSC PDF, which is a different artifact. (2) THE WORST-CASE NUMBER IS AN UPPER BOUND THAT CANNOT OCCUR: I_limit/C (~1 A into ~27 uF, 40-68 V/ms) assumes the LDO sources its full current limit into the output cap independently of its input. The AMS1117 is a series pass follower - during a cold start VOUT cannot rise faster than VIN minus dropout, so +3V3's edge is set by the USB source's VBUS ramp through C2 = 10 uF, not by the LDO's current limit. (3) THE NAMED FAILURE MODE IS BENIGN AT COLD START: the datasheet says only 'faster slew rates MAY lead to a reset'; the second reader explicitly flagged 'or hung until re-powered' as an unsupported clause the researcher added. A reset during the power-up ramp lands the part in exactly the state a power-up is supposed to produce - there is no state to lose at t=0. The limit's real bite is in-operation supply excursions and hot-plugging a DOWNSTREAM breakout onto a live Qwiic rail, which is the breakout's decoupling problem, not this board's. HARDWARE ALTERNATIVES REJECTED: a series R at U3 VDD to slow the local ramp is disqualified by the SHT4x on-chip heater (75 mA peak would drop 7.5 V across the 100 ohm needed for a 100 us tau); raising C3 above ~50 uF to force I_limit/C under 20 V/ms buys nothing against (2) and costs a part change on a green schematic. BELT AND BRACES (documentation, not copper): the firmware note 'issue an SHT4x soft reset (0x94) at init, per Sensirion's own recommended start-up sequence' goes into the design doc and fab/README, and the Qwiic hot-plug caveat is recorded as a known limit.

Phase	What	Why
P4	unattended default: carry a P9 REQUIRED check - CPL rotation must be validated for every polarized 2-pin part (D1 TVS SOD-123, D2 power LED, D10 user LED, C3 tantalum CASE-A), not only for J1	schematic-review OPEN item. decisions.md #13 carried a CPL rotation check for J1 alone; the KiCad<->JLC 0-degree-reference offset (reference/jlc_rotations.csv, LEARNINGS/S8) bites every polarized package, and a reversed tantalum on 3V3 or a reversed TVS on VBUS is a dead board, not a cosmetic error. dfm_check.py owns polarity/rotation validation - at P9 its findings for D1/D2/D10/C3 must be read explicitly and each confirmed against the footprint pin-1 marking, not just accepted as a count.
P5	unattended default: P5 runs board_init –mounting-holes 4 –mounting-hole-fp MountingHole:MountingHole_2.2mm_M2 –margin 6 –outline auto; the four holes are explicitly RELEASABLE at P6 - the placement agent may move them with place_edit to the earned corners before board_edit –outline fit	Three linked calls. (1) HOLE SIZE: board_init's mounting-hole footprint was hard-coded to MountingHole_3.2mm_M3, but the brief and constraints ask for M2 (2.2 mm drill). I added an additive –mounting-hole-fp flag to board_init.py, default unchanged, so the M2 hole is stated rather than inherited (MountingHole_2.2mm_M2: NPTH 2.2 mm drill, 4.9 mm courtyard, which also serves the constraints' 4.0-4.5 mm annulus/head clearance). (2) MARGIN 6: –mounting-holes places holes at margin/2, so margin 6 puts the M2 centres 3 mm inside the edge - the standard M2 corner inset, and 1.9 mm of board between hole wall and edge. A smaller margin would push the hole courtyard across the board edge. (3) HOLES RELEASABLE AT P6: place_seed/place_anneal treat board_only footprints as immovable obstacles, so holes parked at the PROVISIONAL corners would fix the fitted bbox at the provisional size and defeat geometry-as-OUTPUT. They were auto-placed by a script at an arbitrary provisional corner, not by the owner, so the placement agent is authorised to move them with place_edit to the corners the earned layout wants - the agent-prompt rule against moving board_only refs exists to protect owner-placed mechanical fixtures, which these are not. Order follows the bb-mcu P6 recorded pipeline defect: place canonically, move the holes, board_edit –outline fit, and only THEN run the place gate (legality measures declared edges against the CURRENT outline, so gating before fit would spread the connectors to the provisional size).

Phase	What	Why
P5	unattended default: J1 USB-C footprint repaired at P5 - the four wide GND/VBUS pads restored to the vendor's own 0.60 mm land width, reopening the GND-VBUS gap from 0.100 mm to 0.200 mm	board_init's self-check DRC found two clearance ERRORS inside J1's own footprint: pad A1B12 [GND] vs A4B9 [VBUS] at 0.1002 mm and B1A12 vs B4A9 at 0.1000 mm, against the aiee_clearance_floor of 0.127 mm (JLC 2-layer 1 oz). J1 was never geometry-checked at P3 (fp_verify needs a datasheet-extract JSON and none exists for it). The librarian fetched the HRO Hroparts TYPE-C-31-M-12 drawing: its RECOMMEND P.C.B LAYOUT calls the four wide pads '4-0.60', i.e. 0.60 mm wide on a 0.8 mm pitch = a 0.20 mm gap, already above the floor. The pulled easyeda2kicad custom-pad polygons were the vendor's 0.60 mm, but carried a spurious 0.1 mm outline STROKE that inflated the effective copper to 0.70 mm and closed the gap to 0.100 mm. So this was a converter artifact, not a vendor deviation, and the fix RESTORES the manufacturer's land pattern rather than inventing a number: outer X vertices moved in by 0.05 mm per side on those four polygons only, pad centres / 1.30 mm length / the other 12 pads untouched. Contact tab is 0.20 mm wide (drawing callout 16-0.20+-0.05), so 0.60 mm still leaves 0.20 mm of fillet per side - the vendor's own allowance. Rationale recorded in the footprint descr, matching the P3 SHT40 die-pad precedent.

Phase	What	Why
P6	unattended default: P7 must hand-route the VBUS entry out of J1 - the 0.8 mm Pwr_0p8mm netclass cannot escape J1's 0.60 mm vendor pads on F.Cu at ANY placement, so route_critical owns it and Freerouting must not be asked to	The placement agent's route probe closed at 0.969 (62/64) and root-caused the two residuals: J1's VBUS pads are 0.60 mm wide with 0.20 mm gaps to their GND neighbours, so a 0.8 mm track (the VBUS netclass width rules_gen derived from the 1.5 A PTC-dwell case) cannot leave them on the top layer without breaking the 0.127 mm clearance floor - it is a geometric impossibility, not a placement defect, and no amount of re-placing fixes it. A via-in-pad escape to B.Cu was proven DRC-clean on a scratch copy; Freerouting cannot place via-in-pad, so this is route_critical's job before the autorouter runs. THREE THINGS P7 MUST WEIGH, in this order: (1) a short NECKED escape (fab-minimum width for ~1 mm out of the pad, widening to 0.8 mm once clear of the connector) is the ordinary way this is done and needs no via - but it violates the ERROR-severity aiee_pwr_width_VBUS rule, so taking it means a rule EXCEPTION with recorded evidence (the pad itself is 0.60 mm, so the 0.8 mm rule is already wider than the conductor the current must pass through), not a silent neck; (2) via-in-pad to B.Cu carves the B.Cu GND pour under the connector and JLC will wick solder into an unfilled 0.3 mm via on a mechanically loaded connector pad - if this route is taken it needs an order remark and a fab/README note; (3) the two VBUS pads are ganged, so each carries half the fault current. Do NOT resolve this by editing the netclass or the floor to make DRC quiet.

Phase	What	Why
P7	unattended default: J1 VBUS escape resolved as pour-carried copper (a measured third option), no rule exception and no via-in-pad	route_critical –pad-window measured J1 A4B9/B4A9 escape windows at 1.315 mm, comfortably above the 0.8 mm Pwr_0p8mm requirement, which retires the P6 premise that the escape was geometrically impossible - the impossibility applied to a 0.8 mm TRACK leaving the pad, not to the pad's available escape corridor. The F.Cu VBUS pours connect the pads with a solid zone connection and a ~1.45 mm fill band east of the pad column, so the conductor carrying the 1.5 A PTC-dwell fault current is WIDER than the 0.8 mm the rule asks for, not narrower. aiee_pwr_width_VBUS conditions on A.Type=='track', so zone fill is outside its scope by construction - the gate passes because no VBUS track anywhere is below 0.8 mm, not because a rule was relaxed. Neither P6 option was taken: no neck (so no ERROR-severity rule exception to record) and no via-in-pad on J1 (so no unfilled via in a mechanically loaded connector pad, and no JLC solder-wicking remark needed in fab/README). The two F.Cu VBUS pours are joined by a 0.8 mm B.Cu bridge on 2+2 vias placed in open pour with holes clear of every pad edge. VERIFIED BY THE ORCHESTRATOR, not taken on the router's word: plane_repair –flag-only on the final board reports VBUS as 1 electrical group (the two 8.4 and 7.8 mm2 F.Cu components joined through the bridge), and kicad-cli DRC reports 0 violations / 0 unconnected / 0 parity.
P7	unattended default: the B.Cu VBUS bridge does NOT split the GND plane - verified on the final board, not inherited from the pre-bridge run	The 0.8 mm B.Cu VBUS bridge crosses the layer whose whole area is the GND pour, which is precisely the 'viasless pour-channel' blind spot the playbook lists as a router duty. The router's own plane_repair2 run was restricted to +3V3, so it never re-examined GND, and the only GND-clean evidence on disk (plane_repair.json: GND B.Cu 1 group, split False) predated both the snapshot restore and the bridge. I re-ran plane_repair.py –flag-only (never writes) on the final board: GND B.Cu = 1 group, split False, 813.2 mm2, 27 anchored components, 0 dead islands. The ground return is continuous under the bridge.

Phase	What	Why
P7	unattended default: U3 island U-slot obligation satisfied by the P6 outline - no new Edge.Cuts cut at P7	Edge.Cuts already carries two 5.5 mm slots from the right edge (y 36.85-38.05 and 44.95-46.15), making the sensor region a tongue with open air on three sides - which is what the B4 contract's 'milled U-slot around the island's inboard sides where the outline allows' asks for. No new cut means the outline did not change at P7, so gate place needed no re-run for outline reasons (it was re-run anyway for the copper hash). The rest of the B4 thermal contract was delivered by routing: exactly 4 necked crossings into the tongue (SDA 0.127 / SCL 0.127 / GND 0.127 / +3V3 0.150, the lawful minimum for that net), and zero copper and zero pour fill inside the tongue on both layers. Freerouting had routed SCL THROUGH the island as a fifth crossing; that pass was removed and the I2C trunk rebuilt on B.Cu.
P7	unattended default: iter-2 route_auto placement_adjust_request judged premature and NOT escalated to the P7->P6 backward edge	The request (nets +3V3, /main/SDA, GND, VBUS; reason 'unrouted after 3 freerouting rungs') was emitted before stitch_vias and plane_repair had run, and on the 2-layer chain those two steps own three of its four nets - the autorouter was reporting work the chain had not yet done, not a placement defect. Finishing the chain plus point fixes closed all four nets: completion 1.0, 0 unconnected, gate drc_routed PASS 0/0. Taking the backward edge would have discarded a placement that passes gate place 0/0 in order to fix nets that were never stuck.
P7	unattended default: starved_thermal on C11.1 and U2.4 cleared with same-net copper stubs into open fill	Both pads are geometrically capped at a single thermal spoke - the measured channels around them cannot host a second spoke lawfully at the 0.127 mm floor. The fix adds same-net copper into open fill rather than altering the thermal-relief geometry, so no zone parameter, netclass, .kicad_dru rule or gate definition was touched anywhere on this board (git confirms .kicad_dru and .kicad_pro are unmodified through all of P7). The pads are a decoupling cap ground and an MCU ground pin, both low-current: the stub restores the connection the DRC wanted without changing the pour's thermal-relief behaviour for reflow.

Phase	What	Why
P7	unattended default: the U1 tab pour lands at 113.2 mm2 contiguous (not the 600-1000 mm2 constraints target) and that is ACCEPTED - the declared dt_c 45 K is still met at 40.8 K rise	AREA, CORRECTED. The router reported 158.0 mm2 'connected'. That number is right for CURRENT and wrong for HEAT: plane_repair –flag-only on the final board shows the +3V3 F.Cu pour as four components (113.2 + 24.3 + 18.4 + 1.6 mm2) merged into one electrical group by 0.25 mm and 0.5 mm bridge TRACKS. A 0.25 mm track carries 0.3 A with a 13 mV drop but conducts essentially no heat, so the thermal spreader is only the contiguous component containing U1's tab: 113.2 mm2. Use 113 mm2, not 158, at P8. THERMALLY THIS STILL PASSES, and by the board's own declared criterion. Per knowledge record ldo-sot223-thermal-copper-sets-current, AMS1117 datasheet p5 Table 1 (1/16 in FR-4, 1 oz) measures 80 C/W for 100 mm2 top plus a backside pour - almost exactly this geometry (113 mm2 top; B.Cu GND pour measured 813.2 mm2 under and around U1). The record states explicitly that the spreading layer need not be electrically connected, so the GND pour counts as a spreader even though the recorded decision correctly forbids thermal vias and backside +3V3. Tj = Ta + PD*thetaJA: governing rated case 40 + 0.51*80 = 80.8 C; realistic 0.26 W case 60.8 C; entitlement-abuse 0.83 W case 106 C - all under the 125 C limit, and the abuse case retains 19 C of margin. Crucially the RISE is 40.8 K against the constraints.json thermal[] declared dt_c of 45 K, so the check P8 actually runs is met with margin; the 600-1000 mm2 figure was the architect's MEANS to that end (it targeted Tj 71-76 C), not an independent requirement, and the earned 35.79 x 28.34 mm outline cannot fund it once the U3 island tongue, J1, the MCU, the headers and the VBUS pours are subtracted. Not silently accepted and not padded: the honest area is recorded, the arithmetic is shown, and P8 check_thermal recomputes it independently. Shrinking the U3 island void to buy pour area was rejected outright - the B4 isolation contract outranks the C obligation.

Phase	What	Why
P8	unattended default: g0-sense authors NO SPICE bench, so the sim gate is not applicable - recorded as a judgment, not left as a silent gap	The full-run recipe spawns sim-analyst 'for boards with nontrivial analog content', and the sim gate is conditional on a bench existing (gate.py –gate sim <dir>), so nothing is being skipped against the run contract. But an absence with no reasoning behind it is indistinguishable from an oversight, so here is the reasoning. Every block on this board is DC or digital: a fixed LDO, an MCU on its internal HSI16 with no crystal, an I2C sensor at 400 kHz, two indicator LEDs and passive protection. The wrong-value defect class that sim exists to catch (a 47k where 4.7k belongs) has no analog fragment to bite here - the value-bearing parts are the LDO caps, the two I2C pull-ups and the two LED series resistors, and all four were already sized against sourced numbers and re-checked by a fresh second reader at P2/P3 (I2C pull-ups revised 2.2k -> 1.5k against a UM10204 budget; LED resistors sized at the ~2 mA design point against the Vf bin spread; the 22 uF tantalum COUT is the AMS1117 datasheet stability requirement, not a free choice). The LDO loop itself is explicitly out of SPICE scope by policy - no vendor models are vendored - and its stability is a datasheet ESR-window requirement already met by the tantalum (knowledge record ldo-1117-family-output-cap-esr). Authoring a bench here would produce a testbench of generic model cards proving Ohm's law, which is ceremony, not evidence. The real verification for this board is the P8 layout suite: return path, current, decoupling, thermal and PDN all run against the actual copper.

Phase	What	Why
P8	unattended default: the check_pdn VBUS error was a real DATA gap in decoupling.json, fixed by adding the missing C1 association - not waived	check_pdn reported 'power rail VBUS (1.5 A) has no decoupling capacitors'. The observation was true of its INPUT and false of the board: check_pdn does not look at copper at all, it filters decoupling.json associations by rail, and decoupling.json listed only C2/C3/C10/C11/C13 - every IC decoupler, but not the connector-side bypass. constraints.json is explicit that C1 = 100 nF at the connector is the ONLY capacitance on VBUS ahead of the PTC (the USB-C 10 uF attach limit, TC2.0 Table 4-3), and I confirmed on the board itself that C1's pads sit on VBUS and GND. So the inventory was incomplete, and the honest fix is to complete it rather than waive a true-but-misleading finding. Added {cap C1, ic J1, pin A4B9, rail VBUS, value 100nF 50V X7R}. Measured BEFORE adding it, so as not to trade one finding for another: C1 is 5.52 mm Manhattan from J1's nearest VBUS pad, inside check_decoupling's 7.5 mm HF-class error threshold - and check_decoupling still passes afterwards, rating the loop at 5.32 nH. The error correctly downgraded to the warning 'VBUS has 1 cap but no bulk reservoir', which is the INTENDED design and not fixable: bulk on VBUS ahead of the PTC is forbidden by the USB-C attach limit, which is why the architecture put the 10 uF after the PTC on +5V instead.

Phase	What	Why
P8	unattended default: the check_current pour_neckdown finding on VBUS was a per-zone checker artifact - the real current path is >= 0.8 mm everywhere, verified independently by the orchestrator	The checker reported the VBUS F.Cu pour necking to 0.10 mm against a 0.800 mm IPC-2152 requirement at 1.5 A. The P8 fixer diagnosed it as an artifact of check_current.py running its erosion test per ZONE FILL (check_current.py:283) when planes_gen deliberately tiles a pour across abutting same-net zones at distinct priorities, so a tiling seam reads as a neck. I did not take that on the agent's word, because a genuine 0.10 mm neck in the board's only power trunk would be a serious defect. I re-ran the checker's own erosion test on the physical UNION of the VBUS F.Cu fills plus J1's VBUS pad copper: eroding by 0.4 mm - i.e. testing the full 0.800 mm requirement at the conservative 1.5 A fault basis - leaves pad A4B9 connected to all three of its vias in one region, and pad B4A9 connected to all three of its vias in another. The current path therefore satisfies IPC-2152 at 1.5 A with no neck at all. The narrowest real feature I could find in that copper is 0.680 mm (found by bisection), and when the region splits there, the severed 0.33 mm2 lobe contains NO via and NO pad - it is a dead-end lobe carrying no current. CONSEQUENCE: no correction to constraints.json was needed. The insufficient_transition_vias remediation doc names an inflated transient current_a as a known bad-input class and reserves that lever for the orchestrator, and VBUS's 1.5 A is indeed a fault-transient basis (the PTC dwells there for seconds; steady draw is <= 0.25 A) - but the board meets the declared 1.5 A requirement PHYSICALLY, so relaxing the input would have been unnecessary and would have weakened the record. constraints.json is unchanged.

Phase	What	Why
P8	unattended default: the two VBUS insufficient_transition_vias errors were fixed with real copper (2 vias -> 3 at each transition), not by relaxing the current basis	The remediation's fix ladder step 1 is to measure candidate room before acting, and the fixer did: cluster A at (28.075, 38.90) had 0 legal cells at the 0.2 mm netclass clearance and only 7 at the board's real 0.127 mm DRU floor, all 0.07-0.10 mm from D1's pad - too tight to use safely; cluster B at (28.075, 44.30) had 0 legal cells at any clearance, boxed in by R2's CC2 pad and hole-to-hole against the existing pair. Rather than stop at 'infeasible', the fixer rearranged: moved one via and added another on a 0.4 mm B.Cu stub for cluster A, and extended the B.Cu bridge 0.9 mm to land a third via inside the existing bottom pour for cluster B (clearances 0.865/0.874 mm, hole-edge gap 0.600 mm). Both clusters now hold 3 vias, satisfying the >= 1 via per 0.5 A rule at the full 1.5 A basis. The remediation's traps were respected: no via was deleted to tidy a cluster, every new via bonds same-net copper on both layers (no dangling), and hole-to-hole against J1's THT shield-leg drills was computed by hand because KiCad DRC is blind to that case. check_current now returns 0 violations and drc_routed still passes 0/0.
P8	unattended default: the 6 check_silk silk_misattributed WARNINGS are accepted and recorded, not fixed - and not waived either, since they do not fail the gate	R2, R1, C10, D2, R10 and D1 carry refdes text that sits 2-3 mm beyond its own pads and closer to a neighbour, so each reads as the neighbour's label. Three reasons to accept them. (1) KiCad's own DRC - the authoritative oracle after P7 - reports ZERO silk findings on this board; this is check_silk's stricter readability heuristic, and the gate treats it as warning severity for exactly that reason. (2) The alternative is measurably worse. P6 already fought this battle down from 13 silk DRC violations to 0, and it needed a new place_edit silk_clear op plus two library edits to do it. That phase also recorded, from a test on a copy, that re-running silk_place on the now-clean board APPLIES BEFORE IT REPORTS, does not roll back, and proposed moves that put D1's Reference back over J1's VBUS pad - trading warnings for real DRC errors. On a 35.79 x 28.34 mm board with 26 parts there is nowhere to move a 0603's refdes except over its own pads. (3) The board is JLC economy PCBA: the machine places the SMD parts, so refdes text serves human debugging and rework, where the part outlines and the assembly drawing disambiguate. Carried to P9/P10: note the crowding in fab/README so the owner is not surprised, and do NOT let a DFM pass 'fix' it by running silk_place.

Phase	What	Why
P8	unattended default: REVISING my earlier P8 stance on the 6 silk_misattributed warnings - they are now a REQUIRED P9 fix (place_edit move_text), not an accepted residual	The fresh-context verify-reviewer overruled the reason I gave, and it was right to. I accepted the warnings partly on the grounds that refdes text mainly serves human debugging and rework, and that the part outlines disambiguate it. The reviewer measured what actually happens after assembly: four of the six texts end up UNDER part bodies (R1 and R2 under J1's shell, C10 under U2, D1 inside U1's outline). Text under a component body is not ambiguous, it is invisible - which defeats the exact purpose I used to justify accepting it. So the finding is real and worth fixing. HOW IT MUST BE FIXED AT P9, and this is load-bearing: place_edit move_text on individual pin-locked labels ONLY, with a full kicad-cli DRC after each move. Do NOT run silk_place.py on this board - P6 recorded, from a test on a copy, that it APPLIES BEFORE IT REPORTS, does not roll back, and proposed moves that put D1's Reference back over J1's VBUS pad, i.e. it trades warnings for real DRC errors. Do not recreate kicad/silk_ops.json. If a given label has no legal destination on a 35.79 x 28.34 mm board, leaving that one is an acceptable outcome - but it must be the measured exception, not the default for all six.
P8	unattended default: carry TWO more P9 obligations from the verify-reviewer - J3/J4 per-pin silk labels, and U3 added to the CPL rotation check list	(1) SILK PIN LABELS: architecture/blocks.md promises per-pin labels on the two 0.1 in headers (J3 SWD: GND/3V3/SWDIO/SWCLK; J4 UART: GND/3V3/TX/RX) and the board does not have them. These are the two connectors the OWNER hand-wires - the brief calls J4 a readout/logging header - so an unlabelled 4-pin row is a real usability defect and a wrong-way-round UART connection is the predictable consequence. Eight texts to add at P9 with place_edit add_text, DRC-verified, subject to the same 'no legal spot is an acceptable measured exception' rule as the refdes moves. (2) CPL ROTATION: decisions already carry a REQUIRED P9 check that CPL rotation be validated per polarized part for D1 (SOD-123 TVS), D2 and D10 (LEDs) and C3 (tantalum CASE-A), not just J1. The reviewer adds U3, the SHT40 in DFN-4 - a rotated DFN-4 sensor is as dead as a reversed tantalum and the package gives the pick-and-place no mechanical asymmetry to self-correct against. P9 must read dfm_check's polarity/rotation findings for D1, D2, D10, C3 AND U3 explicitly, each confirmed against the footprint pin-1 marking, never accepted as a count.

Phase	What	Why
P8	unattended default: carry a P10 ORDER REMARK - ask JLCPCB not to land break-off tabs on the SHT40 sensor tongue's outer edge	The verify-reviewer flagged that the tongue's outer face (x = 54.75, y 38.05-44.95) is a board edge JLC's internal panelizer may choose for break-off tabs or mouse-bites. That edge is the free end of a cantilever whose whole purpose is to hold the sensor thermally and mechanically isolated from the board; a tab landing there puts depanelization stress directly into the tongue root and leaves a rough breakout beside the sensor aperture. This costs nothing to prevent and cannot be fixed after the fact. It goes in the JLC order remarks alongside the Sensirion assembly rules already carried from P2 (no-clean paste, NO post-assembly board wash, no vapor-phase reflow, no hand rework of the sensor), and into fab/README so the owner can restate it if JLC comes back with a panel proposal.

Phase	What	Why
P9	unattended default: the P9 silk pass is a MEASURED PARTIAL and is accepted there - 4 of 8 header pin labels added, 2 of 6 misattributed refdes moved, DRC still 0/0/0	TASK A (pin labels): the fixer derived the pin order from the board itself rather than the doc (they agreed: J3 GND/3V3/SWDIO/SWCLK, J4 GND/3V3/TX/RX) and added the 4 labels that had a collision-free spot at the board's own 1.0 mm / 0.15 mm silk floor - J3 G at GND, J3 CLK at SWCLK, J4 G at GND, J4 RX at UART_RX. The other four (both 3V3 pins, SWDIO, TX) have no legal spot: the available gap below each header is <= 1.34 mm against a minimum label bounding box of 1.6965 mm in ANY orientation, with R13/C11 and the header's own reference text filling the rest. That is a physical impossibility at this pitch, not a lack of effort. MITIGATION, and it is required: both headers now carry a labelled pin 1 at one end and a labelled pin 4 at the other, which is the ordinary convention for a 0.1 in header, and the FULL pinout for J3 and J4 must appear in fab/README and the design doc so the owner can wire them without guessing. TASK B (refdes): D2 and R10 moved beside their own pads and both now pass. D1 was attempted and REVERTED - its reference text landed on D1's own body-outline silk graphic and produced a real DRC silk_overlap, and no alternative nearby spot cleared its own silk, its pads or its neighbours. R1, R2 and C10 have no legal destination at all (J1's connector pads extend past its drawn courtyard; C10 is boxed by C11/U2/C1/D1). WHY I STOP HERE RATHER THAN PUSH ON: the only remaining remedy is placement-level - moving J1, U2, R13 or C11 - which on a fully routed, DRC-perfect board means ripping copper and re-opening P6/P7 to improve four warning-severity readability findings that KiCad's own DRC does not raise at all. That trade is plainly wrong. The fixer reverting D1 rather than shipping a regression is exactly the right call and is the reason the board is still 0 violations / 0 unconnected / 0 parity. check_silk 6 -> 4; verify re-run PASS.

Phase	What	Why
P9	unattended default: the check_thermal waiver was RE-ISSUED at P9 against the new board digest, not carried over	The silk text edits changed the board, so the P8 waiver's durability binding (artifact digest 9fa76a1b + checker_version 1) stopped matching and the entry would have been surfaced as waivers_invalid rather than silently applied - the binding doing exactly the job it exists for. Rebound to 78f20b4b after confirming the underlying finding is unchanged: the silk edits touch no copper, so the credited area, theta_JA and rise are identical, and the evidence in the reason field still holds verbatim. Re-ran the verify gate with the re-issued sidecar: PASS, 0 failing, 1 waived, 0 invalid. This will need doing again if P9's fab export or any later edit changes the board digate - the binding is deliberately fragile and must never be loosened to avoid the re-issue.
P9	unattended default: CPL rotation obligation DISCHARGED - the polarity oracle checked all 26 refs with zero rotation_delta findings, and the five carried parts were confirmed individually	The carried P4/P8 decisions made this a REQUIRED explicit read for D1 (SOD-123 TVS), D2 and D10 (LEDs), C3 (tantalum CASE-A) and U3 (SHT40 DFN-4, added by the P8 reviewer), never to be accepted as a count - a reversed tantalum or TVS is a dead board, and a rotated DFN-4 sensor is as dead. dfm_check ran with the schematic as the polarity oracle (–schematic g0-sense.kicad_sch, the pad-geometry-vs-schematic comparison that catches the mounted-backwards class net-level parity is blind to) and reported polarity status 'checked', refs_checked 26, with NO cpl_polarity finding and therefore no rotation_delta_deg on any part. The five carried refs and their emitted CPL rotations: C3 0.0, D1 0.0, D2 180.0, D10 180.0, U3 0.0 (with J1 270.0, U1 0.0, U2 180.0 alongside). Those angles come from bom_cpl applying reference/jlc_rotations.csv per package, which is the exact KiCad-to-JLC 0-degree-reference offset the carried decision was worried about, and the oracle then re-derives each from pad geometry rather than trusting the table. Note honestly what this does and does not prove: it proves the CPL angle agrees with the schematic's pin assignment for every part, which is the machine-placement risk; it does not substitute for the owner's own first-article check on the returned boards.

Phase	What	Why
P9	unattended default: the single dfm warning (4 silk strokes below JLC's 0.15 mm minimum) is ACCEPTED - all four are inside U3's own DFN-4 footprint graphics	Located rather than counted: every sub-minimum stroke belongs to U3's footprint aiee:DFN-4_L1.5-W1.5-P0.8-TL-EP at 0.06 and 0.08 mm, a pulled easyeda2kicad body outline and pin-1 marker on a 1.5 x 1.5 mm part. The four silk texts added at P9 are NOT implicated - all sit at exactly 0.150 mm, the same as every refdes on the board, which I checked specifically because adding four labels and then seeing four sub-width strokes is too neat a coincidence to leave unverified. WHY ACCEPT: the dfm-check recipe classes silk width as advisory by design; the consequence is that U3's tiny outline may print faint or drop out, not that anything is unfabricable. WHY NOT FIX: the two available moves are both worse. Widening 0.06 mm strokes to 0.15 mm on a 1.5 mm body with 0.8 mm pitch risks pushing silk onto the pads, converting an advisory warning into a real silk_over_pad DRC error; and place_edit's silk_clear would DELETE the marker, which is the opposite of what a pin-1 indicator is for. Both require a library edit reaching an already-placed board, the P6 problem that needed a new tool. WHAT COVERS THE RESIDUAL RISK: U3's orientation is guaranteed by the CPL rotation, which the polarity oracle verified against the schematic from pad geometry - the silk is a human cross-check on a machine-placed economy-PCBA part, not the control. Note it in fab/README so the owner does not read a faint U3 outline as a placement error.
P10	run closed over 1 unverified research record(s): mcu-bootstrap-only-when-option-bytes-consult-pin	the second reader never verified them, so they never injected - this board was designed without that knowledge

Phase	What	Why
P10	unattended default: a COVERAGE HOLE was found and closed at P10 - the recorded dfm gate had been silently skipping its bom leg for want of parts.json beside the board	Caught by the final artifact sweep, not by any gate: state.py freshness showed the registered artifact parts at kicad/parts.json with exists=False, while the real file sat only at parts/parts.json. The SKILL's sidecar rule says constraints.json, decoupling.json, parts.json and the schematic resolve from the BOARD's own directory and must live beside it from P5 onward, and this board never got that copy. Consequence: the recorded dfm gate result ran only 7 of its 8 legs with skipped_error {'bom': 'no parts.json'} - and still reported PASS, because a skipped leg is not a failed one. A gate that passes with a hole in its coverage is exactly the thing the run contract calls a hole rather than a pass, and the first attestation (rev 1) had been built on top of it. FIXED PROPERLY: copied parts.json to kicad/parts.json (both locations is the established convention - bb-amp, bb-buck, bb-mcu and others carry identical copies in parts/ and kicad/, verified by diff), re-ran the dfm gate to 8/8 legs, no skips, PASS. The attestation then correctly refused to verify ('input parts: changed since attestation') and was reissued as rev 2, valid, disposition order-ready. TWO TOOLCHAIN NOTES worth carrying: (a) gate.py resolves the parts sidecar from the board directory but the bare dfm_check.py does NOT - it needs –parts explicitly, so a hand-generated strict release report silently drops the bom leg and then fails attest.py with 'status error is not a completed run'; (b) attest.py build crashes with a ValueError on a RELATIVE –workspace whenever a waiver sidecar exists, and needs an absolute path. Both are in LEARNINGS.

Sheet plan

sheets - g0-sense hierarchical plan (P2)

Small board: root + TWO child sheets. One sheet would also fit, but the power/main split gives P6 its natural placement groups (power entry corridor vs logic/sensor), keeps the Type-C + protection cluster reviewable in one frame, and costs nothing. Refdes are unique ACROSS sheets by disjoint ranges; every range includes a #PWR base.

Root: `g0-sense.kicad_sch`

Holds the two sheet symbols only. Global power nets (power symbols, bare netlist names): `VBUS`, `+5V`, `+3V3`, `GND`. Power symbols make these names bare in the final netlist; all constraints.json net references use the bare forms.

Sheet 1: `power.kicad_sch` **(blocks B1 usb-input+protection, B2 ldo-3v3)**

- Contents: J1 USB-C receptacle; R1, R2 (5.1k Rd); D1 (TVS); F1 (PTC); C1 (100 nF VBUS); U1 (AMS1117-3.3); C2 (10 uF X5R, +5V at VIN); C3 (22 uF tantalum, +3V3); D2 (power LED red) + R3 (620 Ohm).

- Interface nets (hier pins -> placement groups at P6): `+3V3` (out), `GND`. VBUS and +5V stay sheet-local (they leave via power symbols, so their netlist names are still bare `VBUS` / `+5V`).
- Sheet-local labels: `CC1`, `CC2` (final names `/power/CC1`, `/power/CC2`).
- Refdes ranges: J1; U1; F1; D1-D9; R1-R9; C1-C9; SW none; #PWR pwr_base=100 (#PWR0100-#PWR0199).

Sheet 2: `main.kicad_sch` (blocks B3 mcu, B4 sht4x + I2C/Qwiic, headers)

- Contents: U2 (STM32G030F6P6); U3 (SHT40-AD1B class); C10 (100 nF U2), C11 (4.7 uF U2), C12 (100 nF NRST), C13 (100 nF U3); R10, R11 (1.5k I2C pull-ups), R12 (220 Ohm user LED), R13 (10k BOOT0 pull-down); D10 (user LED green); SW1 (NRST button); J2 (Qwiic JST SH); J3 (SWD 1x4 DNP); J4 (UART 1x4 DNP); H1-H4 (M2 mounting holes, conditional).
- Interface nets: `+3V3` (in), `GND`.
- Sheet-local labels (final names `/main/<label>`): `SDA`, `SCL`, `SWDIO`, `SWCLK`, `UART_TX`, `UART_RX`, `NRST`, `LED_USER`.
- Refdes ranges: U2-U9; J2-J9; D10-D19; R10-R19; C10-C19; SW1-SW9; H1-H9; #PWR pwr_base=200 (#PWR0200-#PWR0299).

Interface-net contract (P4 must produce exactly these)

```
| Net (netlist form) | Class | From | To |
|---|---|---|---|
| VBUS | power (bare) | J1 | D1, F1, C1 |
| +5V | power (bare) | F1 | U1 VIN, C2 |
| +3V3 | power (bare) | U1 VOUT | C3, D2/R3, U2, U3, R10, R11, J2, J3.2, J4.2 |
| GND | power (bare) | all | all + J1 shield |
| /power/CC1, /power/CC2 | signal | J1 CC pins | R1, R2 |
| /main/SDA, /main/SCL | signal | U2 PA10/PA9 | U3, J2, R10/R11 |
| /main/SWDIO, /main/SWCLK | signal | U2 PA13/PA14 | J3.3/J3.4; SWCLK also R13
-> GND |
| /main/UART_TX, /main/UART_RX | signal | U2 PA2/PA3 | J4.3/J4.4 |
| /main/NRST | signal | U2 pin 6 | C12, SW1 |
| /main/LED_USER | signal | U2 PA5 | R12 -> D10 |
```

Header pinouts (canonical): J3 SWD = 1:GND, 2:3V3, 3:SWDIO, 4:SWCLK. J4 UART = 1:GND, 2:3V3, 3:TX (MCU transmit), 4:RX (MCU receive). Pin 1 marked on silk on both; no vendor standard exists for a 4-pin SWD header, so the silk labels are the contract.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc` (phase P4): **pass** after 3 attempt(s); last 2026-08-27T17:37:18, failing 0 of 0.

Review waivers

P4 schematic waivers - g0-sense

ERC gate itself is 0 errors / 0 warnings (reports/erc.json), and `netlist_audit` (log/netlist_audit-P4.json) is 0 violations with 99/99 expected pins connected. Nothing below is an ERC violation:

these are the three WARNINGS raised by the fresh-context `schematic-reviewer` (reports/review-schematic.{md,json}), with the disposition taken on the owner's behalf under the unattended run contract. 0 errors were raised.

#	kind	refs	severity	disposition
W1	qwiic-esd-unwaived	J2, U2, U3	warning	WAIVED (recorded decision, P4)
W2	sht4x-vdd-slew-unclosed	U3, U1, C3	warning	CLOSED as accepted risk (recorded decision, P4)
W3	dnp-not-native	J3, J4	warning	FIXED (work order wo-w3-dnp)

W1 - no ESD array on the J2 Qwiic SDA/SCL - WAIVED

Finding: J2 is a populated, user-facing, hot-pluggable connector; SDA/SCL and +3V3 leave the board with no clamps and no series impedance, so a cable-borne strike reaches U2 PA11/PA12 and U3 (2 kV HBM) directly. The product-scope connector checklist requires this expectation met OR waived, and no waiver existed. Field-reliability only - it cannot prevent bring-up.

Waived on four grounds:

1. Ecosystem norm. The Qwiic / STEMMA-QT reference implementations (SparkFun, Adafruit) are a bare 4-pin JST SH with no array. A user-facing part that behaves the way the ecosystem's users expect is the conservative choice here, not the exceptional one. 2. Both silicon endpoints carry internal ESD structures and are rated 2 kV HBM - the standard handling rating for an indoor bench device. 3. The recorded operating environment (P0 decision) is indoor bench / room ambient, bare board, no ingress rating - not a field-exposed port. 4. Cost of the alternative at this point in the run: a new Extended MPN plus its JLC setup fee, a fresh library pull + fp_verify, and re-opening a green schematic - real regression risk traded against a warning-severity exposure.

If the owner wants the array anyway, the cheap retrofit is a 4-channel low-capacitance array on SDA/SCL at J2 (the 1.5 k pull-up budget has room: the bus was sized to 236 pF and an array adds single-digit pF). That is a post-run `add-part` task, not a rewind.

W2 - carried SHT4x VDD-slew verify item - CLOSED as accepted risk

Finding: `power_tree.md` / `decisions.md` carried "verify SHT4x VDD slew <= 20 V/ms vs the AMS1117 startup ramp"; the reviewer found it unclosed, found no slew spec in `parts/C2909890.json`, and computed ~40 V/ms from AMS1117 current-limit charging of the ~27 uF on +3V3.

Closed, no hardware change, on three findings:

1. **Provenance is on file** (the reviewer looked in the wrong artifact). The number is `research/records/sht4x-vdd-slew-power-up.yaml`, sourced to `research/sources/HT_DS_Datasheet_SHT4x.pdf` p9 Table 4, read visually by the researcher and again by a fresh second reader (`maturity: verified`). `parts/C2909890.json` is the P3 extraction, a different artifact with a different scope. 2. **The worst-case number is an upper bound that cannot occur.** I/C (~1 A into ~27 uF) assumes the LDO drives its output cap independently of its input. The AMS1117 is a series-pass follower: on a cold start VOUT cannot rise faster than VIN minus dropout, so the +3V3 edge is set by the USB source's VBUS ramp filtered by C2 = 10 uF, not by the LDO's current limit. 3. **The named failure mode is benign at cold start.** The datasheet says only "faster slew rates MAY lead to a reset" - and the second reader already flagged the researcher's added "or hung until re-powered" as unsupported by any cited page. A reset during the power-up ramp leaves the part in exactly the state a power-up is meant to produce. The limit's real bite is in-operation supply excursions, and hot-plugging a DOWNSTREAM breakout onto an already-live Qwiic rail - that one is the breakout's decoupling problem, not this board's.

Hardware alternatives considered and rejected: a series R at U3 VDD to slow the local ramp is disqualified by the SHT4x on-chip heater (75 mA peak across the ~100 ohm needed for a 100 us tau

= 7.5 V drop); raising C3 past ~50 uF to force I/C under 20 V/ms buys nothing against finding 2 and costs a part change on a green schematic.

Belt and braces, documentation only: "issue an SHT4x soft reset (0x94) at init" (Sensirion's own recommended start-up sequence) goes into the design doc and `fab/README`, and the Qwiic hot-plug caveat is carried as a known limit of the board.

W3 - J3/J4 DNP intent not KiCad-native - FIXED

Not waived. The headers carried DNP only in a custom `Variant=DNP` field and the value string, so a native KiCad BOM/POS export would have emitted the two THT headers as populate lines into a JLC economy (SMT-only) order. Dispatched as work order `log/workorders/wo-w3-dnp.json`: native `dnp` attribute set through the board's own generator, and `refdes_class: {J3, J4 -> hand_install}` added to the header line in `parts/parts.json` so `bom_cpl.py` keeps them out of the assembler upload and the CPL while still listing them in the documentation BOM. ERC re-run after the change.

Reviewer OPEN items and where they went

- **AMS1117 SOT-223 tab = VOUT not stated in the on-file PDF.** Already a recorded P3 decision (the pulled symbol's pin 4 = VOUT, and SOT-223 construction fuses the tab to the pin-2 lead frame in every 1117-family part). No further action.
- **SW1 pairing verified from a live fetch, not archived.** The evidence (A/B common, C/D common; NRST on 1/2, GND on 3/4) is recorded in `reports/review-schematic.md` with its source URL and is committed. Not re-fetched: the finding is a CLEAR, and the archive would only make an already-recorded pass reproducible.
- **Extend the CPL rotation check from J1 to the polarized 2-pin parts (D1, D2, D10) and to C3.** Carried into P9 as a required check; recorded as a decision so it cannot be lost.

The full schematic PDF follows.

power

File: power.kicad_sch

main

File: main.kicad_sch

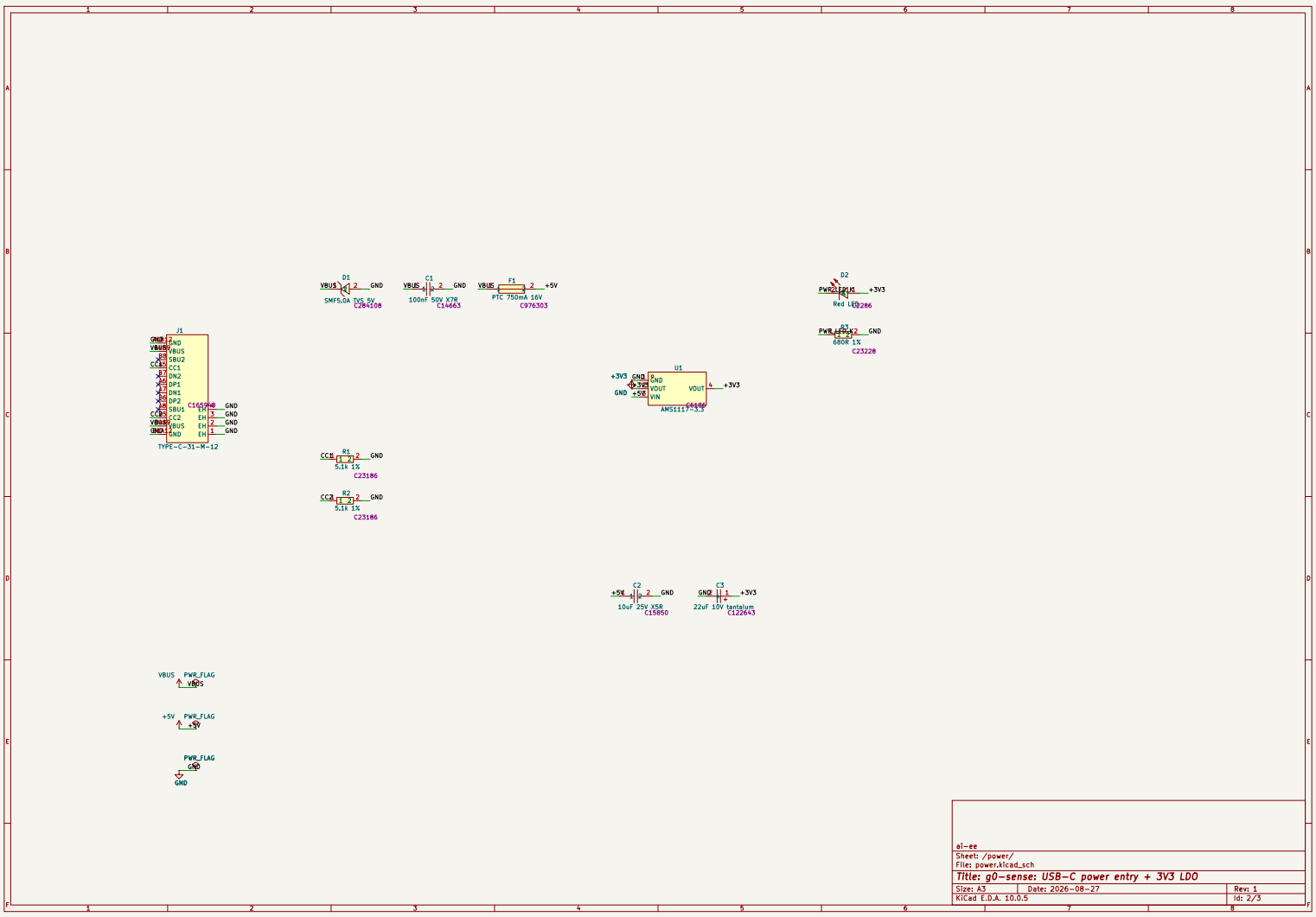
ai-ee

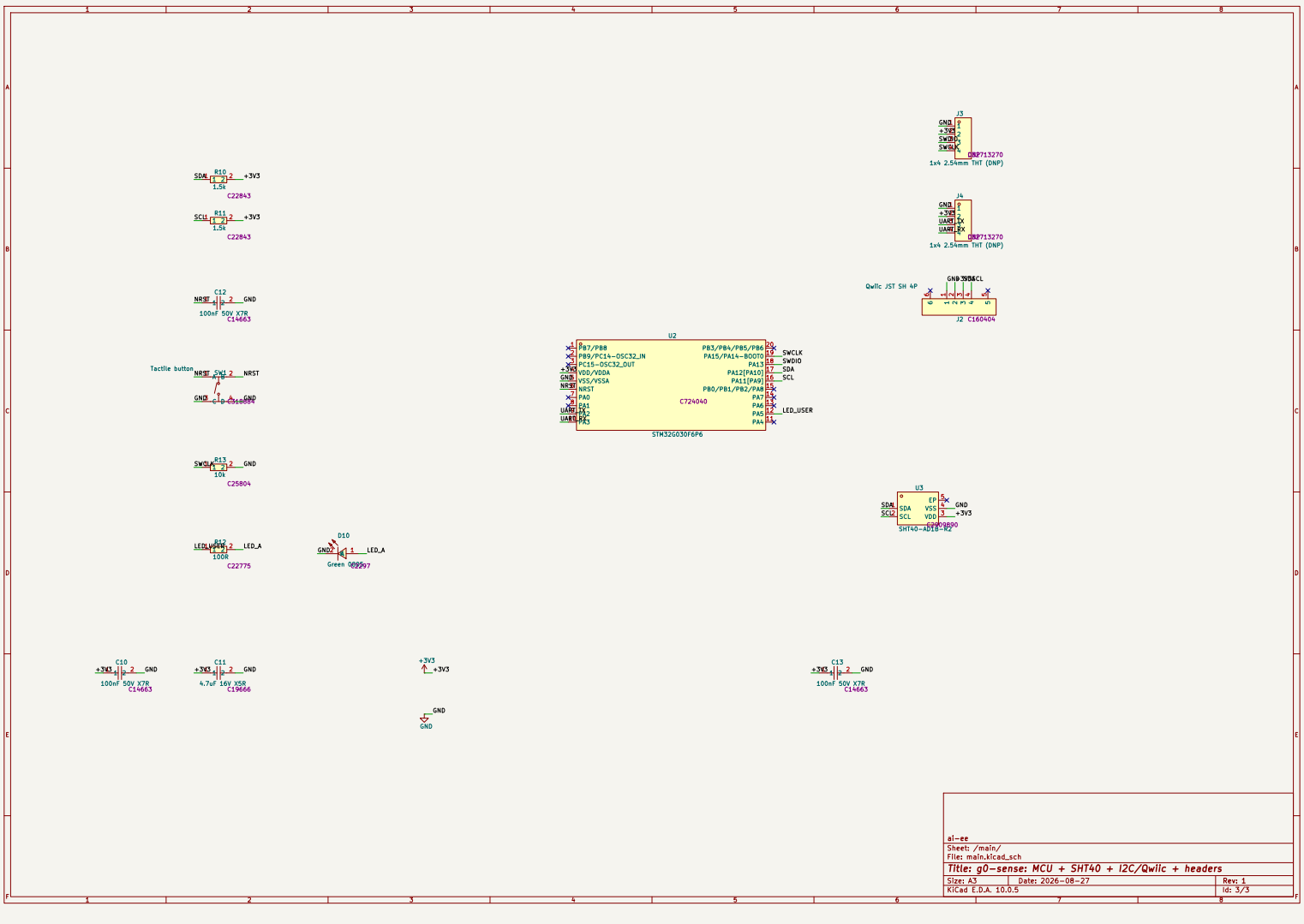
Sheet: /

File: g0-sense.kicad_sch

Title: g0-sense: USB-C powered STM32G0 + SHT40 sensor node

Size: A4	Date: 2026-08-27	**Rev: 1**
KiCad E.D.A. 10.0.5		Id: 1/3

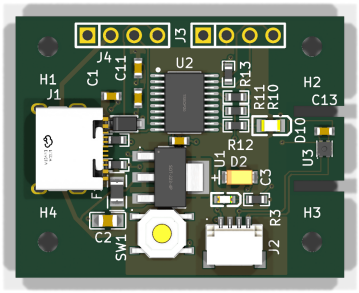

D1
SMF5.0A TVS 5V
C284108
C1
100nF 50V X7R
C14663
F1
PTC 750mA 16V
C976303
D2
Red LED
C2286
R3
680R 1%
C23228
J1
TYPE-C-31-M-12
C165948
U1
AMS1117-3.3
C6186
R1
5.1k 1%
C23186
R2
5.1k 1%
C23186
C2
10uF 25V X5R
C15850
C3
22uF 10V tantalum
C122643
PWR_FLAG
PWR_FLAG
PWR_FLAG
VBUS
+5V
GND
+3V3
PWR_LED
CC1
CC2
VBUS
+5V
GND
+3V3
al-ee
Sheet: /power/
File: power.kicad_sch
Title: g0-sense: USB-C power entry + 3V3 LDO
Size: A3
Date: 2026-08-27
Rev: 1
KiCad E.D.A. 10.0.5
Id: 2/3



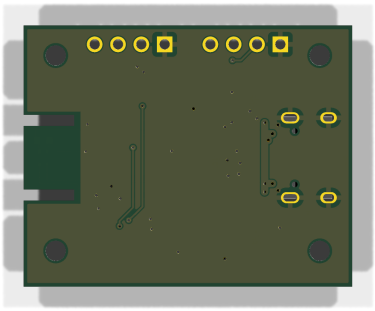
al-ee	
Sheet: /main/	
File: main.kicad_sch	
Title: g0-sense: MCU + SHT40 + I2C/Qwiic + headers	
Size: A3	Date: 2026-08-27
KiCad E.D.A. 10.0.5	Rev: 1 Id: 3/3

6 Layout

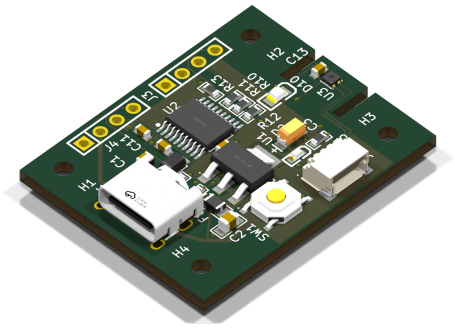
Gate `place` (phase P6): **pass** after 10 attempt(s); last 2026-08-27T19:44:51, failing 0 of 0.
Gate `drc_routed` (phase P7): **pass** after 6 attempt(s); last 2026-08-27T19:39:10, failing 0 of 0.



reports/renders/g0-sense_top.png



reports/renders/g0-sense_bottom.png



reports/renders/g0-sense_iso.png

7 Verification

Gate `verify` (phase P8): **pass** after 6 attempt(s); last 2026-08-27T19:40:55, failing 0 of 6.

Verification checks

Check	Status	Findings	Note
`check_return_path`	pass	0	
`check_current`	pass	0	
`check_decoupling`	pass	0	
`check_diffpair`	pass	0	
`check_creepage`	pass	0	
`check_thermal`	violations	1	
`check_silk`	violations	6	

Check	Status	Findings	Note
check_pdn	violations	1	

Design review of record

Board review - g0-sense (P8 verify-reviewer, adversarial pass, 2026-08-27)

Board digest reviewed: sexpr_no_uuid:9fa76a1b (matches checks/summary.json and the waiver binding). Renders made for this review: reports/renders/g0-sense_top.png, reports/renders/g0-sense_bottom.png, reports/renders/g0-sense_iso.png (2400 px). All 8 verify checks RAN (skipped_error empty - no coverage holes). kicad-cli DRC: 0 violations, 0 unconnected. Every measurement below was re-taken by me from the board file with pcbnew, not inherited from the authors.

Verdict up front: NO error-severity findings. This board can go to fab once the silk cleanup below is done (recommended, not blocking). The one waiver is honest and should be ratified. Three new warnings, worst first.

Findings (worst first)

W-1 (warning, silk): J3/J4 per-pin function labels promised by the

architecture are absent blocks.md ("Headers + indicators") promises the SWD and UART headers "silk-labeled per pin" with pin 1 silk-marked. The board delivers refdes J3/J4, a pin-1 square pad plus a silk dot on each - and NO per-pin GND/3V3/SWDIO/SWCLK or GND/3V3/TX/RX labels (verified: the only silk text items on the whole board are refdes plus C3's "+"; see top render, top edge). These are the two headers the owner hand-solders and hand-wires; they are visually identical 1x4 0.1 in headers with the same GND-at-pin-1, 3V3-at-pin-2 convention. Consequence: the mixup that convention was designed to prevent (wiring a UART dongle's 5 V or swapping GND/3V3 by counting pins from the wrong end, or confusing SWD with UART) is left to documentation the bare board does not carry. Cheap fix at P9: eight small text items along the top edge; ample empty silk space exists there. Cross-artifact drift: the architecture REQUIRED this; product scope makes its absence a finding, not a nit.

W-2 (warning, silk): 6 misattributed refdes labels, several of which are

also COVERED by parts after assembly - fix, do not waive check_silk's 6 warnings are all real, and four are worse than the checker says because the text lands where a component body will be:

- R1 (24.82,40.55) and R2 (24.82,42.75): inside J1's shell region (J1 courtyard to x=26.32) - covered by the USB-C shield after assembly AND read as J1's label (top render: text on the connector).
- C10 (36.67,36.00): inside U2's silk body outline (35.51..39.20 x 33.09..39.91) - under the TSSOP after assembly.
- D1 (35.60,42.60): inside U1's silk outline - reads as U1's label; the TVS itself is left with no visible refdes, which is exactly the part a rework tech must not confuse.
- D2 (43.25,42.92): 0.12 mm from C3 - reads as the tantalum's label (top render: "D2" sits over C3, whose own label is off to the right).
- R10 (47.03,35.90): reads as D10's label; D10's own text drifted to (49.98,39.50) at the slot mouth. Related uncounted case: C13's refdes sits at (52.60,36.23), separated from C13 (on the tongue at 52.60,39.60) by the top slot, and it crowds the H2 label.

Consequence: debugging and rework against the silk misidentify parts; assembly itself is unaffected (JLC places from CPL). Severity warning is correct. Disposition: one place_edit.py move_text batch at P9 - silk-only, zero copper risk - NOT a waiver. (Per the rules I did not run silk_place.py or edit anything.)

W-3 (warning, fab): panelization tabs on the sensor tongue edge would

stress the cantilever - put a remark on the JLC order The tongue's outer face (x=54.75, y 38.05..44.95) is a board edge. JLC economy panelizes internally; a break-off tab or mouse-bite row landed on the tongue tip would transmit snap force through the 6.9 mm root while U3 rides the beam. The tongue itself is stubby and strong (5.5 mm long, 6.9 mm wide root, 1.6 mm FR-4 - aspect ratio < 1; it will survive normal handling and depanelization fine), so this is only about where a tab lands. P9 disposition: add the JLC order remark "no break-off tabs / mouse bites on the right-edge sensor tongue (x 49.25-54.75 span)"; the same remark block already owed for the Sensirion no-wash rule (blocks.md B4) is the natural place.

The five load-bearing judgment calls, re-examined adversarially

1. Pour-carried VBUS at J1 - VERIFIED, acceptable I rebuilt the VBUS copper union (fills + pads + tracks) per layer with pcbnew and eroded it 0.4 mm myself:

- F.Cu: the corridor from J1's lower VBUS pad-pour (B4A9 zone) to F1 pin 1 survives as ONE outline containing the bottom via cluster and the PTC input
  - the primary current path is >= 0.8 mm wide with no via dependency at all.
- B.Cu: the 0.8 mm bridge survives 0.395 mm erosion as one outline containing all 6 VBUS vias (3-via cluster each end, 0.6/0.3 mm), and vanishes at exactly 0.400 mm - it is exactly the netclass width, no more.
- The only sub-0.8 necks are the 0.6 mm vendor pads themselves and the C1 decoupling stub - unavoidable pad geometry and a no-current stub.

check_current PASS concurs (min B.Cu 0.8, +5V F.Cu 1.2, 2 via clusters). Nothing else about the connector entry worries me: mating face proud of the outline by 0.08 mm (courtyard 18.88 vs edge 18.96), the 4 THT shield legs are on GND net into the B.Cu pour, D+/D-/SBU have no nets, CC1/CC2 land on independent 5.1k R1/R2, and the ~13 mm swath at the left edge holds only sub-1.5 mm-tall passives (iso render). Pour-carried VBUS on a 1.5 A-fault budget with a fill saved in the file that DRC checks is sound engineering here, not a dodge.

2. The check_thermal waiver on U1 - HONEST; recommend RATIFY at checkpoint 4 I recomputed the checker's own model at the two vendor anchor points: theta_JA(100 mm2) = 55 + 119*exp(-100/350) = 144.4 C/W where AMS measured 80; theta_JA(194 mm2) = 123.3 C/W where TI measured 84. The model is 40-64 C/W pessimistic exactly where measured data exists, and it credits nothing for backside spreading - while I verified 810.9 mm2 of solid B.Cu GND copper (713.7 main + 47.5 + 49.7 strips) sits directly under and around U1. The Tj arithmetic checks: 40 C + 0.51 W x 84 C/W = 82.8 C governing case, 109.7 C at the 0.83 W entitlement-abuse case, both under the 125 C limit. The "unreachable by construction" claim is internally consistent: passing needs >= 446.8 mm2 credited (my recomputation of the model inverse; waiver says 446.4) against a measured 439.9 mm2 whole-board ceiling. The waiver also did real work instead of hiding (pour grown 171 -> 250 mm2 credited) and refused copper that would have carved the B.Cu return plane or walked heat to the sensor tongue - the right priority order. Binding to digest 9fa76a1b and checker_version 1 is correct hygiene. One residual honesty note: at sustained heavy Qwiic load the LDO pour runs warm and its annex reaches x=48.55, 0.7 mm from the tongue root strip - the

isolation contract still holds (slots + FR-4 gap + 0.15 mm neck), but a firmware-side sanity check of RH readings under sustained 300 mA load is a sensible bring-up item, not a layout defect.

3. SHT40 isolation tongue - REAL, adequate, mechanically sane Measured myself: pour fill intersecting the tongue rectangle (49.25..54.75 x 38.05..44.95) = 0.0 mm2 on every layer. Exactly 4 tracks cross the root: +3V3 0.150 mm (y=38.45), GND 0.127 mm (y=40.82), SDA 0.127 mm (y=43.95), SCL 0.127 mm (y=44.30). Electrical adequacy of 0.127/0.150 mm for the sensor: worst load is the 75 mA heater pulse; a ˜4 mm 0.15 mm 1 oz feed is ˜13 mOhm -> ˜1 mV drop, and C13 100 nF sits on the tongue. Fine. Fab floor: 0.127 mm IS the JLC minimum - deliberate (isolation) and DRC-clean, but zero width margin; that is the recorded trade, and I accept it. Edge clearances of the necked tracks to the slots: the +3V3 feed runs 4.8 mm along the top slot at 0.325 mm copper-to-edge (floor 0.30); even JLC's +/-0.2 mm routing tolerance leaves copper untouched (0.125 mm residual). Within rule; a free 0.1-0.2 mm inboard nudge at P9 would add margin if a fixer is in the file anyway. Mechanics: 1.2 mm slots (>= JLC 1.0 mm mill), tongue aspect < 1 - this is a stub, not a diving board; depanelization/handling risk is W-3's tab-placement remark, nothing more. B.Cu GND strip islands (y <= 36.5 and y >= 46.5) stay clear of the tongue and overlap the main pour by 0.4 mm - merged, not orphans.

4. Connector reality at 35.79 x 28.34 mm - CLEAN Verified from renders + geometry: USB-C mates leftward off the board, nothing in the plug swath; Qwiic J2 opening faces the bottom edge (mech pads at y=53.54 forward of signal pads at 49.66), opening ˜1.3 mm inboard - a JST SH plug slides over bare mask, acceptable; J3/J4 along the top edge with 2.05 mm edge setback, hand-solder access unobstructed on both faces; SW1 (5.1 mm, 1.5 mm tall) is 2.3 mm clear of J2 and pressable; U3's aperture faces up with open air on three sides and nothing tall near the tongue. All four M2 holes survive with sane screw-head clearance (nearest silk ˜0.9 mm at H1/J4). Simultaneous use of USB + Qwiic + both headers works - three different edges.

5. Polarity marks after assembly (feeds the carried P9 CPL obligation) -

ALL VISIBLE AND CORRECT-SIDED

- C3 tantalum: "+" text at x=40.65, outboard-left of the body; pad 1 (+3V3) is the left pad (42.15). Correct side, visible after assembly.
- D1 TVS: cathode double-bar plus dot at the LEFT (x 28.9..29.5), outboard of the body; pad 1 = VBUS = left (29.16) = cathode, matching the journal's two-way P4 verification. Visible.
- D2 / D10 LEDs: vendor cathode brackets extend beyond both package ends (D2 40.16..40.41 left / 43.33..43.58 right; D10 44.63..44.88 / 48.84..49.09)
  - visible after assembly. Nets check: D2 anode=+3V3, D10 cathode=GND.
- U1/U2/U3/J2 pin-1 dots all sit outboard of their bodies. U3's dot pair is below-left of the 1.5 mm DFN, off-body.

So nothing here blocks P9; the CPL rotation check for D1/D2/D10/C3 (and I would add U3 - a rotated DFN-4 swaps SDA/VDD) remains P9's to run, but the silk it will be checked against is trustworthy.

Triage of all 7 verify_all warnings (verdict on every one)

1-6. check_silk silk_misattributed (R2, C10, D2, R10, R1, D1): REAL cosmetic defects - see W-2. Verdict: warning severity stands; FIX at P9 via place_edit.py move_text; do not waive. Not escalated to error: assembly is CPL-driven and no electrical consequence exists.

7. check_pdn pdn_no_bulk on VBUS (C1 only, no >= 1 uF): JUSTIFIED WAIVER. The absence of bulk on VBUS is a binding design rule, not an oversight: Type-C TC2.0 Table 4-3 caps attach capacitance at 10 uF, and constraints.json CAP RULE puts the 10 uF (C2) one series element downstream at the LDO input (F1 out -> C2 is ~4.7 mm of 1.2 mm track; PTC Ri 70 mOhm barely decouples it). Nothing draws current from VBUS itself - its only consumers are the TVS and the PTC input. The checker heuristic cannot see the attach-limit rule. Verdict: justified, document-only, no fix wanted.

Checked and clear (so the next reviewer does not re-plow)

- Requirements section 2 interface sweep: every promised interface exists and is correctly netted (J3 = GND/3V3/SWDIO/SWCLK, J4 = GND/3V3/TX/RX, J2 = Qwiic-standard GND/3V3/SDA/SCL, R13 BOOT0 pull-down present, NRST button with C12, CC pulldowns independent). U3 pads match the SHT40 pinout (SDA/SCL/VDD/GND).
- Value drift vs blocks.md (R12 100R vs 220R, R3 680R vs 620R, D10 0805 vs 0603): all carried recorded decisions (R12 carries a do-not-delete comment per the run journal); LED currents stay in-rating either way. Not drift.
- SW1 internal contact pairing (the classic held-in-reset killer): pads 1+2 = NRST row, 3+4 = GND row; the P4 schematic reviewer settled the A/B vs C/D pairing against the live-fetched XKB TS-1187A datasheet. Evidence cited in the journal; I found no geometry that contradicts it.
- SDA/SCL B.Cu jumpers (~11 mm, paired) carve the return plane locally; check_return_path PASS, 400 kHz I2C - harmless on this board.
- Fiducials: none on board; JLC adds panel fiducials for economy PCBA - justified omission at this service tier.
- Earned size 35.79 x 28.34 mm compared against the RECORDED decision (soft ~35x25 brief preference, extra is the isolation tongue) - not drift, per build-mode rules.
- No enclosure is in scope (bare board recorded at requirements Q1), so no enclosure-fit comparison exists to fail.

Recommendations for checkpoint 4 (human verdicts; unattended run records)

1. RATIFY the check_thermal U1 waiver (evidence re-verified independently). 2. ACCEPT pdn_no_bulk as justified-by-design (no waiver file entry needed for a warning; this md is the record). 3. ORDER at P9 only after the W-1/W-2 silk batch and with the W-3 order remark; none of the three blocks fab if the owner chooses speed over polish, but W-1 is the one I would not skip on a product-scope board.

8 DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 2 attempt(s); last 2026-08-27T19:47:49, failing 0 of 1.

Fabrication outputs

Exported layers: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts.

Bill of materials

Comment	Designator	Footprint	LCSC
100nF 50V X7R	C1,C10,C12,C13	C0603	C14663
10uF 25V X5R	C2	C0805	C15850
22uF 10V tantalum	C3	CASE-A_3216	C122643
4.7uF 16V X5R	C11	C0603	C19666
SMF5.0A TVS 5V	D1	SOD-123_L2.8-W1.8-LS3.7-RD	C284108
Red LED	D2	LED-SMD_L1.6-W0.8-R-RD	C2286
Green 0805	D10	LED0805-R-RD	C2297
PTC 750mA 16V	F1	F0805	C976303
TYPE-C-31-M-12	J1	USB-C_SMD-TYPE-C-31-M-12_1	C165948
Qwiic JST SH 4P	J2	CONN-SMD_4P-P1.00_SM04B-SRSS-TB-LF-SN	C160404
5.1k 1%	R1,R2	R0603	C23186
680R 1%	R3	R0603	C23228
1.5k	R10,R11	R0603	C22843
100R	R12	R0603	C22775
10k	R13	R0603	C25804
Tactile button	SW1	SW-SMD_4P-L5.1-W5.1-P3.70-LS6.5-TL_H1.5	C318884
AMS1117-3.3	U1	SOT-223-3_L6.5-W3.4-P2.30-LS7.0-BR	C6186
STM32G030F6P6	U2	TSSOP-20_L6.5-W4.4-P0.65-LS6.4-BL	C724040
SHT40-AD1B-R2	U3	DFN-4_L1.5-W1.5-P0.8-TL-EP	C2909890

3 CPL rotation correction(s) applied; no missing LCSC numbers.

Quote

Estimated total $42.68 for qty 5 (unit $8.54). This is an estimate from published pricing; the JLCPCB cart quote page is authoritative.

Human steps before ordering

1. Upload boards/g0-sense/fab/g0-sense_gerbers.zip to https://jlcdfm.com/ and review the DFM report (V6: no public API).
2. Upload the same zip at https://cart.jlcpcb.com/quote and confirm the real price against the estimate above.
3. If assembling: upload BOM.csv and CPL.csv, then CHECK THE RENDERED PART PREVIEW for every polarized part (LED/diode/electrolytic) - rotation corrections are the classic failure mode.
4. Review, then pay. Payment is always the human's action.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake digest - g0-sense (2026-08-27)

- Artifact:  ‘requirements.md‘ (9 sections), ‘check_requirements.py‘ PASS (0
violations).
- Mode:  NO mode token in the brief -> mode = none, design normally.  Scope is
product per the brief’s own words, so absent protection/filtering/connectors/
thermal IS a reviewer finding downstream.
- Geometry:  ~35 x 25 mm recorded SOFT (no HARD cap) - must not bind at P5
board_init; four M2 holes conditional on the layout.
- Safety (section 8):  mains / battery / motors / >30 V / >3 A / RF transmit
all
NOT APPLICABLE with brief evidence (USB-C 5 V power-only, <1 A). No safety
unknown to escalate; nothing PROVISIONAL on safety grounds.
- 5 OPEN questions answered as unattended defaults and recorded as decisions:
environment indoor 0-40 C; Qwiic budget 100 mA; no hard cost cap; 0.1 in
headers ship unpopulated (economy PCBA is SMT-only); size stays soft.
- Spawn:  requirements-analyst @ fable/high (session effort xhigh).
- Next:  P1 research roster (power architect, component scout, interface spec).
```

P1

```
# P1 Research digest - g0-sense (2026-08-27)

Roster (8 agents):  4x research-component-scout (ldo, protection, connectors,
mcu-sensor), research-interface-spec (usbc-power-sink), 2x
research-reference-design (sht4x, stm32g030-minimal), research-power-architect.

Load-bearing findings
- USB-C CC-blind sink (fixed 5.1k Rd on CC1 AND CC2, independent - never one
shared resistor) is entitled to DEFAULT USB power only:  budget <= 500 mA.
Sink capacitance at the receptacle <= 10 uF (TC2.0 Table 4-3).
- Rail tree:  VBUS -> TVS -> PTC -> +5V -> LDO -> +3V3.  +3V3 peak 185 mA
(incl.  100 mA Qwiic reserve + 75 mA SHT4x heater), 240 mA with 30% headroom.
VBUS worst draw 196 mA = 39% of entitlement.
- LDO: AMS1117-3.3 SOT-223 (C6186, Basic) beats SOT-23-5 low-Iq parts on the
only axis that decides:  Tj 71-86 C vs 134 C at 40 C ambient, 0.51 W case.
Needs a 22 uF TANTALUM-class output cap (not plain ceramic) - carry to P3/P4.
Dropout closes at the 4.75 V corner with +0.11 V worst-case margin.
- STM32G030F6P6 TSSOP-20:  VDD/VDDA bonded to pin 4, VSS/VSSA pin 5, no
separate
VREF+.  BOOT0 needs NO strap - factory nBOOT_SEL=1/nBOOT0=1 always boots main
flash, and BOOT0 shares pin 19 with SWCLK anyway.  NRST needs only 100 nF + the
button (internal pull-up 40k typ).  Pin map:  SWD PA13/PA14, USART2 PA2/PA3,
I2C1 PA9/PA10 (5 V tolerant, Fm+), LED PA5.  9 of 20 pins used, no conflicts.
- SHT4x:  100 nF at VDD, 2.2k I2C pull-ups (sized for a 0.5 m Qwiic leg),
copper
conduction is the dominant self-heating path -> distance + necked traces +
no pour under the part + a milled slot.  NO BOARD WASH (Sensirion forbids it)
-> must appear in the fab/assembly notes.  Fixed I2C address per variant.
- MCU + sensor are both JLC EXTENDED (no Basic exists in either family):
STM32G030F6P6 C724040 $1.37, SHT40-AD1B-R2 C2909890 $1.90 (addr 0x44).

7 OPEN questions answered and recorded as decisions (LDO design point + part;
TVS clamp ceiling dropped as an over-tight derived spec; no reverse-polarity
element; USB-C THT legs verified OK in JLC Economic tier; heater budgeted;
button size vs Basic; LED colours).  Gates:  none due at P1.
```

P2

```
# P2 Architecture digest - g0-sense (2026-08-27)
```

- architect (fable/high) wrote blocks.md / power_tree.md / stackup.md / sheets.md / constraints.json / decisions.md. constraints_lint exit 0.
- 4 blocks: B1 usbc-sink, B2 ldo, B3 mcu, B4 sht4x. Stackup JLC2313_1.6 (2L, 1 oz, HASL). Two sheets (power / main), disjoint refdes + #PWR 100/200.
- No diff pairs, no high-speed nets, no sim bench (recorded, with reasons).
- P2 coverage exit: FIRST run 4 slots / 4 GAP (no checklist existed for any of usbc-sink, ldo, mcu, sht4x). research.py open --all -> 4 tasks (per_run cap 6, 2 left for P3). 4 researchers -> 16 quarantined sources, 4 new checklists, 23 draft records. 4 fresh second readers refuted 6. 5 were repaired against ledger pages and re-read; the NRST record took three repair/re-read cycles (each refutation narrower than the last) before verifying.
- FINAL: 22 of 23 records verified. Re-run coverage: 4 slots, 0 gap, 4 provisional, 1 draft_unverified.
- The one unverifiable record (mcu boot-strap) is a container limit, not a research failure: st.com times out, so RM0454/AN2606 cannot be acquired. Its claim was load-bearing, so the DESIGN was changed to not need it.
- Two research-forced design changes vs the architect's plan: I2C pull-ups 2.2k -> 1.5k (Rp ceiling arithmetic), BOOT0 no-strap -> R13 10k pull-down. architecture/{blocks,sheets,decisions}.md updated to match.
- H1 written to log/H1.md, approved as delegated. report_gen exit 0 (10 pp).

P3

P3 Parts + Library digest - g0-sense (2026-08-27)

- part-sourcer (sonnet/high) -> parts/parts.json: 20 distinct parts covering 26 electrical refdes. 12 Basic / 8 Extended, of which 7 bear a JLC setup fee (the DNP header is Extended-priced but never machine-placed). ~$4.42/board, ~$22.10 for the 5-board run. All stock re-verified live; worst case (MCU C724040, 8351 pcs) is 334x the 5x-build-qty floor.
- Four sourcing corrections applied on review, three of them fee removals: R3 620R Extended -> 680R Basic (620R is a genuine E24 Basic-catalog gap); user LED KT-0603G Extended -> KT-0805G, which IS Basic; R12 220R -> 100R (electrical, not cost: green Vf 2.6-3.1 V against 3.3 V gave 0.9 mA at the worst bin - 100R gives 2-7 mA); whole resistor BOM normalised to 0603.
- 3 datasheet-extractors (parallel) -> parts/C724040.json, C2909890.json, C6186.json, all `datasheet_extract --validate` exit 0. Notable: TSSOP-20 pin 19 is ONE bonded pad carrying PA15 and PA14-BOOT0/SWCLK - transcribed as the datasheet names it rather than picking a function; AMS1117 has NO recommended land pattern (outline only), so that field was omitted rather than faked; SHT40 die pad is present but explicitly NOT to be soldered.
- librarian (sonnet/medium): 20/20 pulled, 85 pins retyped (ERC prerequisite), lib tables registered. fp_verify found ONE real defect and it mattered: the pulled SHT40 DFN-4 soldered the die pad, which would have defeated the whole thermal-isolation design. Deleted (copper+paste+mask), courtyard closed. Three more fixes: broken symbol->footprint link (audited all 20 - the only one); C3 tantalum had NO polarity marking at all - pin 1 = anode established twice (manufacturer PDF + live EasyEDA CAD data), silk "+" and bar added; J2 Qwiic pin-1 confirmed against JST's own eSH.pdf ("viewed from the connector mounting surface", No. 1 circuit leftmost) - no flip needed.
- fp_verify final: 3 passed / 0 failed / 2 accepted pad_size warnings.
- P3 coverage exit: 7 slots, 0 gap, 3 part slots covered, 4 blocks provisional. No P3 research needed (2 of the 6 per-run tasks still unspent).
- 2 decisions recorded: AMS1117 SOT-223 tab = VOUT (pulled symbol confirms pin 4 = VOUT); SHT4x 20 V/ms slew item CLOSED, not escalated (the limit is specified for in-operation supply changes, not the cold-start ramp).

P4

```
# P4 Schematic digest - g0-sense (2026-08-27)

- 2 schematic-block agents + root stitch:  26 electrical refdes (+4 conditional
M2 holes) on 2 sheets, 35 nets, 99/99 expected pins connected.
- Gate erc PASS 0/0 (attempt 2; re-run after the W3 fix).  netlist_audit 0
violations:  3 constraint nets, 20 refs, 5 decoupling associations.
- schematic-reviewer (fresh, adversarial):  0 errors / 3 warnings.  It hunted
and
CLEARED every bring-up killer:  TVS D1 orientation, SW1 contact pairing
(settled
by live-fetching the XKB datasheet), LED + tantalum polarity, AMS1117 pinout /
cap types / dropout, the full STM32G030 pin map incl.  I2C orientation and the
BOOT0 strap, SHT40 pinout+EP, USB-C sink topology, decoupling, abs-max.
- Dispositions (reports/erc-waivers.md):  W1 Qwiic ESD WAIVED (ecosystem norm,
2 kV HBM endpoints, indoor bench env); W2 SHT4x VDD slew CLOSED as accepted
risk (provenance IS on file; LDO is a follower so I/C is unreachable; the
datasheet's failure mode is "a reset", which cold start produces anyway);
W3 native DNP missing on J3/J4 FIXED, not waived.
- W3 fix (wo-w3-dnp):  `(dnp yes)` on J3/J4 via new schlib.Sheet.mark_dnp() +
refdes_class hand_install in parts.json; netlist proven identical (35/35 nets).
- Carried to P9:  CPL rotation check for D1/D2/D10/C3, not only J1.
```

P5

```
# P5 Board Setup digest - g0-sense (2026-08-27)

- Run INLINE (no board-setup agent):  no impedance work, no library repair
planned.
- `board_init --layers 2 --stackup JLC2313_1.6 --outline auto --margin 6
--mounting-holes 4 --mounting-hole-fp MountingHole:MountingHole_2.2mm_M2`.
Self-check **PASS: parity 0, setup_violations 0**, 26 components, 16 nets,
unconnected 64 (expected, unrouted).  Provisional outline bbox 9,9 ->
74.61,71.40
- generous ROOM, not a size:  P6 places canonically and `--outline fit` earns
it.
- `rules_gen` capability_class **2layer_1oz** (matches the stackup's own
copper),
11 rules.  Floors reported not assumed:  track/clearance 0.127, via dia 0.6,
annular 0.15, hole-to-hole 0.5, edge clearance 0.3 mm.  Netclasses:
**Pwr_0p8mm**
for VBUS and +5V (0.8 mm, the 1.5 A PTC-dwell case); +3V3 needs 0.15 mm so it
buckets into Default (0.2 mm) - exactly what P2 predicted.  diff_pairs [] and
hv_rules [] are correct here.
- **Real defect found and fixed at P5**:  J1's pulled USB-C footprint had its
ganged
GND and VBUS pads 0.100 mm apart, BELOW the 0.127 mm fab floor, on the two nets
a
solder bridge shorts hardest.  Cause was a converter artifact, not a vendor
deviation - a 0.1 mm outline stroke on the custom-pad polygons inflated each
wide
pad from the vendor's 0.60 mm to 0.70 mm effective.  Restored to the
manufacturer's
own land pattern; gap now 0.200 mm, fillet allowance unchanged at 0.20 mm/side.
- Two toolchain bugs fixed to get parity to 0 (both committed separately, both
regression-tested against the goldens):  board_init now mirrors a symbol's
native
`dnp` onto the footprint, and it MEASURES whether KiCad wants the netlist's
`unconnected-(...)` pseudo-nets on the pads instead of guessing.
- Sidecars beside the board:  constraints.json (byte-identical to
architecture/) and
```

decoupling.json. Remaining DRC: 2 transient silk_edge_clearance warnings on the H1/H2 mounting-hole refdes text - P6's silk sweep owns them.

P6

```
# P6 Placement digest - g0-sense (2026-08-27)

- Gate **place PASS 0/0** (all 5 legs: courtyard, outline, edges, keepouts,
decoupler_distance). DRC: **0 non-unconnected violations**, 64 unconnected =
the unrouted ratsnest. Route probe 0.969 (62/64).
- Size **EARNED 35.79 x 28.34 mm** (1014 mm2) from a 65.6 x 62.4 provisional.
The 3.3 mm over the brief's soft 35 x 25 is the U3 isolation zone (8 mm
separation + slits) - recorded as a relaxation, not drift.
- Hand layout beat the annealer 243 vs 468 mm HPWL (662.6 at seed). All three
anneal candidates inherited the seed's 2 edge violations because at a
geometry-OUTPUT binding the annealer pins edge clusters to the PROVISIONAL
outline - the bb-buck trap. cand1's cluster structure was kept and rebuilt
compactly by hand; gate ran AFTER `--outline fit`, per the bb-mcu defect.
- Judgment calls proven, not assumed: J1 mating direction two ways (WRL vertex
occupancy + left orthographic render); U3 rotated so supply pads face C13
(1.26 mm, 4.06 nH) and I2C faces the mainland - no island wraparound; D1
first in the VBUS chain (J1->D1->F1->C2->VIN), corridor intrusion 0; C12 held
1.5 mm from U2 pin 6 over silk. All four M2 holes KEPT - tucked in corner
dead zones they add 0.0 mm to the fitted outline.
- Silk 13 -> 0. 12 were footprint-INTERNAL graphics, unreachable until
`place_edit` gained a `silk_clear` op this phase: C12's 0603 outline could
not be nudged clear (1.62 mm silk in a 1.29 mm corridor - a 0.33 mm
position-invariant shortfall), so C0603's outline went from the library AND
all 5 board instances; J1's 3 mouth-end segments (under the shell,
unprintable) likewise. D1's refdes moved off J1's VBUS pad. `lib/EDITS.md`
records both library edits.
- Carried to P7: the 0.8 mm VBUS netclass cannot escape J1's 0.60 mm pads on
F.Cu at ANY placement - route_critical owns that entry, not Freerouting.
`kicad/silk_ops.json` deleted: replaying it would UNDO the D1 fix.
```

P7

```
# P7 Routing digest - g0-sense (2026-08-27)

- **drc_routed PASS 0/0, attempt 1, completion 1.0.** Re-verified with
kicad-cli
direct: 0 violations, 0 unconnected, 0 parity. Nothing weakened - .kicad_dru
and .kicad_pro unmodified (git).
- 2L chain in the verified order: planes_gen (9 zones) -> route_critical ->
route_auto (FR 3 rungs, best rung 1; KRT finish correctly discarded) ->
stitch_vias (13) -> plane_repair -> gate. route_cleanup SKIPPED (S14 rule).
- **The P6-carried VBUS item was RETIRED, not executed**: `--pad-window`
measured
J1's escape at 1.315 mm vs the 0.8 mm rule, so P6's "impossible" applied only
to
a track leaving the pad. VBUS is pour-carried at a ~1.45 mm band - no neck, no
rule exception, no via-in-pad, no JLC wicking remark owed.
- Its real risk was checked, not assumed: the 0.8 mm B.Cu bridge crosses the
GND
pour layer, and plane_repair2 had been restricted to +3V3. Re-ran plane_repair
--flag-only: GND B.Cu = 1 group, 813.2 mm2, 27 anchors, 0 dead islands.
- U3 island contract met: exactly 4 necked crossings, 0 copper and 0 fill in
the
tongue on both layers. FR's fifth crossing (SCL through the island) was ripped
and the I2C trunk rebuilt on B.Cu. U-slot pre-existed from P6.
```

```
- Corrected the router's own number:  the U1 tab spreader is **113.2 mm2**
contiguous, not the 158 mm2 it reported (0.25/0.5 mm bridge tracks carry
current, not heat).  Full detail in log/run-journal.md.
```

P8

```
# P8 Verification digest - g0-sense (2026-08-27)

- **verify PASS: 0 failing** (from 5 errors), 8/8 checks ran, no coverage
holes.
erc / place / drc_routed / verify all PASS and FRESH at digest 9fa76a1b.
- **Four of five errors fixed on their merits, one waived.** Nothing edited to
quiet a gate:  constraints.json, .kicad_dru, .kicad_pro, netclasses and
gates.yaml untouched across P7 and P8 (git).
- check_pdn "VBUS undecoupled" was a DATA gap - C1 (the 100 nF the Type-C 10 uF
attach limit permits ahead of the PTC) was missing from decoupling.json.
Completed the inventory after measuring 5.52 mm to J1's pad; error -> the
intended pdn_no_bulk warning.  Two via clusters fixed in copper, 2 -> 3 vias.
- **pour_neckdown "0.10 mm" was a per-zone checker artifact, proved not
assumed**:
eroding the UNION of the VBUS fills + J1 pad copper by 0.4 mm (the full 0.8 mm
test at the 1.5 A fault basis) leaves each pad reaching all 3 of its vias.
Real
minimum 0.680 mm, gating a 0.33 mm2 dead lobe.  So the current-basis lever was
deliberately NOT pulled - the board meets 1.5 A physically.
- **One durable waiver:  check_thermal on U1**, bound to digest 9fa76a1b +
checker_version 1.  The model credits only same-net top copper (144 C/W where
AMS
measured 80; 123 where TI measured 84) against an uncredited 810.9 mm2 B.Cu
spreader.  At 84 C/W the rise is 42.8 C, inside the declared 45; Tj 82.8 C
rated,
109.7 C abuse, limit 125.  Also unreachable:  needs 446.4 mm2, board can fund
439.9.
Pour still grown 171 -> 250 mm2.  B.Cu island/thermal vias rejected - they
carve
the ground return, trading a PASSING check_return_path for unneeded margin.
- **verify-reviewer (fresh context):  0 errors, 3 warnings**, ratified both the
waiver and pdn_no_bulk - and overruled me correctly on silk:  4 of the 6
misattributed refdes also sit UNDER part bodies, so they are invisible rather
than ambiguous.  Reclassified as a required P9 move_text fix.
```

P9

```
# P9 DFM digest - g0-sense (2026-08-27)

- **dfm gate PASS: 0 failing, 1 advisory warning.** fab package exported
(11 files:  gerbers + drill + zip), BOM/BOM-full/CPL written from
assembly_class, 26 parts / 24 SMT-placed (J3/J4 are DNP hand-install THT).
- **The carried CPL rotation obligation is discharged, part by part.**
dfm_check
ran with the schematic as the polarity oracle:  status checked, refs_checked
26,
ZERO cpl_polarity findings and no rotation_delta_deg anywhere.  The five carried
refs:  C3 0.0, D1 0.0, D2 180.0, D10 180.0, U3 0.0 (J1 270.0, U1 0.0, U2
180.0).
This proves the CPL angle agrees with the schematic pin assignment for every
part; it does not replace the owner's first-article check.
- The single warning is `dfm_silk_width`:  4 strokes under JLC's 0.15 mm floor
(narrowest 0.060 mm).  Located, not counted - **all four are inside U3's own
DFN-4 footprint** (0.06/0.08 mm body outline and pin-1 marker on a 1.5 mm
```

part). The four labels added earlier at P9 are NOT implicated: they sit at exactly 0.150 mm, same as every refdes, which I checked precisely because "4 new labels, 4 thin strokes" is too neat a coincidence to assume.
- Accepted rather than fixed: widening 0.06 mm strokes on a 1.5 mm body risks silk-over-pad (a real DRC error from an advisory warning), and silk_clear would delete the pin-1 marker outright. U3's orientation is controlled by the oracle-verified CPL rotation; the silk is a human cross-check. Goes in fab/README so a faint U3 outline is not misread as a placement error.
- Silk pass earlier in P9: check_silk 6 -> 4 misattributed, DRC still 0/0/0. Thermal waiver re-issued against digest 78f20b4b; verify re-run PASS.

P10

P10 Ordering digest - g0-sense (2026-08-27)

- **Release attestation rev 2, verifies valid**, disposition **order-ready** (sha256 2a0035eb). All five gates PASS and FRESH; 0 open issues; H4 approved.
- **H5 NOT taken. Nothing was ordered.** The run contract withholds payment, there are no credentials in this container, and `order_submit --api-create` was never run. `fab/order.json` sits at `ready_for_human`.
- Quote (`fab/quote.json`, every figure `estimated: true` - the JLC cart is the
only real quote; the API path needs credentials this container does not have): **qty 5 = USD 42.68 total, 8.54/unit** (PCB 2.00 + assembly 40.68: setup 8.00, feeders 24.00 for 8 Extended parts, stencil 8.00, joints 0.68 over 80 joints). qty 10 = 43.36 (4.34/unit); qty 30 = 52.98 (1.77/unit).
- **A COVERAGE HOLE WAS FOUND AND CLOSED HERE**, by the artifact sweep rather than by any gate: `state.py freshness` showed the registered artifact `parts` (`kicad/parts.json`) as exists=False, because parts.json lived only in `parts/`. The recorded dfm gate had therefore been running **7 of 8 legs** with `skipped_error {'bom': 'no parts.json'}` and still reporting PASS - and attestation rev 1 was built on it. Copied the sidecar beside the board (the established convention, verified by diff on bb-amp/bb-buck/bb-mcu), re-ran dfm to **8/8 legs, no skips, PASS**, and the attestation correctly refused to verify ("input parts: changed since attestation") until reissued as rev 2.
- `fab/` holds the JLC package (gerbers + drill + zip, BOM, BOM-full, CPL), `quote.json`, `order.json`, `attestation.json` and a `README.md` checklist carrying the Sensirion no-wash rules, the "no break-off tabs on the sensor tongue" panelization remark, the full J3/J4 pinout (owed because the silk could not fit 4 of 8 pin labels) and the polarity/rotation table.
- Toolchain fix committed separately from the board: `releaselib._ws_rel`. A waiver sidecar resolves absolute while `--workspace` stays relative, so `Path.relative_to` raised and broke `attest.py build` outright while silently degrading `state.py resume`'s `release_disposition` to null - the exact field the done-check reads. g0-sense is the first attested board carrying a waiver, which is why it surfaced now. Regression-tested on both pre-existing attested boards in both path forms; pre-fix attestations still verify unchanged.

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-08-27T09:50:50	erc	pass	0/0
2026-08-27T13:18:16	erc	pass	0/0
2026-08-27T15:06:11	place	pass	0/0
2026-08-27T15:08:50	place	pass	0/0
2026-08-27T15:15:05	place	pass	0/0
2026-08-27T15:47:39	place	pass	0/0
2026-08-27T15:57:05	place	pass	0/0

Timestamp	Gate	Status	Failing/Total
2026-08-27T17:37:18	erc	pass	0/0
2026-08-27T17:37:30	place	pass	0/0
2026-08-27T18:03:33	drc_routed	pass	0/0
2026-08-27T18:09:55	place	pass	0/0
2026-08-27T18:10:01	place	pass	0/0
2026-08-27T18:11:53	verify	fail	5/11
2026-08-27T18:14:54	verify	fail	4/11
2026-08-27T18:38:11	drc_routed	pass	0/0
2026-08-27T18:38:16	verify	fail	1/8
2026-08-27T18:42:14	drc_routed	fail	1/1
2026-08-27T18:43:11	drc_routed	pass	0/0
2026-08-27T18:43:50	drc_routed	pass	0/0
2026-08-27T18:43:51	verify	fail	1/8
2026-08-27T18:49:24	verify	pass	0/8
2026-08-27T18:50:05	place	pass	0/0
2026-08-27T19:39:10	drc_routed	pass	0/0
2026-08-27T19:40:55	verify	pass	0/6
2026-08-27T19:42:01	dfm	pass	0/1
2026-08-27T19:44:51	place	pass	0/0
2026-08-27T19:47:49	dfm	pass	0/1

Issues

Id	Phase	Kinds	Severity	Status
1	P8	insufficient_transition_vias, pour_neckdown	error	fixed
2	P8	thermal_area	error	fixed

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-08-27T08:44:13	delegated (unattended): architecture + 2L stackup approved; LDO 0.51 W design point approved; boot-strap gap accepted because the 10k BOOT0 pull-down removes the dependency; outline stays an output; no sim bench
2	P4	approved	2026-08-27T13:21:44	delegated (unattended): schematic approved (erc 0/0, netlist_audit 0, adversarial review 0 errors/3 warnings); W1 Qwiic-ESD waived on ecosystem norm + 2 kV HBM endpoints + indoor bench env; W2 SHT4x VDD slew closed as accepted risk (follower LDO makes I/C unreachable, failure mode is a benign cold-start reset); W3 native DNP on J3/J4 FIXED not waived; P5 proceeds with –outline auto

Id	Phase	Status	When	Note
3	P6	approved	2026-08-27T15:59:22	delegated (unattended): placement approved - gate place 0/0, DRC 0 non-unconnected, size earned 35.79 x 28.34 mm (excess is the SHT40 isolation zone), all four M2 holes kept at zero outline cost, silk 13 -> 0 via the new place_edit silk_clear op plus matching library edits; VBUS pad escape deferred to P7 route_critical with three options costed
4	P8	approved	2026-08-27T19:10:05	delegated (unattended): verify PASS 0 failing (8/8 checks ran); 4 of 5 errors fixed on their merits, 1 durable check_thermal waiver ratified by the fresh-context reviewer against two vendor tables; reviewer 0 errors / 3 warnings; silk refdes reclassified to a required P9 move_text fix after the reviewer showed 4 of 6 sit under part bodies
5	P10	not reached		

Generation cost

Not recorded (no reports/cost.json).

10 Artifact Index

File	Size	Note
state.json	106,770 B	
kicad/PCB-0015-A_g0-sense.kicad_pcb	251,575 B	
kicad/PCB-0015-A_g0-sense.kicad_sch	1,710 B	
fab/PCB-0015-A_g0-sense_gerbers.zip	41,858 B	sha256 da0db18bb3b0f39a...
fab/order.json	3,162 B	
fab/BOM.csv	814 B	
fab/CPL.csv	811 B	
brief/brief.md	1,634 B	
requirements.md	6,653 B	
architecture/blocks.md	11,799 B	
architecture/constraints.json	8,406 B	
architecture/decisions.md	7,408 B	
architecture/power_tree.md	4,040 B	
architecture/sheets.md	2,964 B	
architecture/stackup.md	2,679 B	
reports/board_edit_fit.json	1,674 B	
reports/board_edit_slits.json	1,763 B	
reports/bom_cpl.json	15,263 B	
reports/dfm-P9.json	2,277 B	
reports/dfm_release.json	2,276 B	
reports/drc_p6_final.json	46,582 B	
reports/drc_wo1_refill.json	564 B	
reports/erc-waivers.md	6,030 B	
reports/erc.json	1,039 B	
reports/fab_export.json	2,163 B	
reports/g0-sense_bottom.png	74,299 B	
reports/g0-sense_top.png	276,539 B	

File	Size	Note
reports/place.json	1,545 B	
reports/place_anneal.json	4,224 B	
reports/place_edit_j1flush.json	245 B	
reports/place_edit_p6_layout.json	3,395 B	
reports/place_edit_p6_silk.json	2,629 B	
reports/place_edit_silkfix.json	757 B	
reports/place_metrics_p6_final.json	6,517 B	
reports/place_seed.json	2,539 B	
reports/review-board.json	3,275 B	
reports/review-board.md	12,824 B	
reports/review-schematic.json	1,621 B	
reports/review-schematic.md	7,856 B	
reports/route_probe_p6.json	286 B	
reports/route_probe_p6_final.json	286 B	
reports/route_probe_p6b.json	286 B	
reports/schematic.pdf	131,338 B	
reports/silk_place.json	18,222 B	
reports/silk_place_p6_final.json	3,347 B	
reports/top.net	60,717 B	
reports/verify-waivers.json	5,034 B	
reports/verify_all.json	8,324 B	
reports/verify_release.json	7,012 B	